

# Stranded capacity: measuring and validating the gap between state digital provision and citizen reach

## Abstract

The Human Development Index (HDI) remains the most widely used composite measure of national well-being, yet its health–education–income structure predates the internet and omits the digital foundations of contemporary human capability. Existing digital extensions treat state e-government capacity and citizen digital reach as substitutable, averaging the gap between them into invisibility. This paper develops a neutrosophic decomposition that holds the two apart. The E-Government Development Index is separated into a truth component (measured state capacity), a falsity component (citizen non-penetration), and a bipolar indeterminacy component (the signed divergence between its own service and infrastructure sub-indices). Truth and falsity combine conjunctively to give stranded capacity  $\Phi$ , the share of nominal provision lying beyond citizen reach; the poles of the indeterminacy component give its configuration. The framework is applied to 184 economies, and the reach-adjusted score is carried into an augmented HDI on the 158 economies with complete data. Four findings emerge. First,  $\Phi$  is externally validated: conditional on state capacity and income, it predicts lower citizen uptake of digital government services in demand-side household survey data ( $-0.462$ ,  $p = 2.3 \times 10^{-5}$ ,  $N = 145$ ), while showing no association with digital payment activity in general — so it captures state-mediated uptake specifically rather than connectivity. Second, facade intensity is concentrated in sub-Saharan Africa and Southern Asia and survives an income control, and no economy of the Global North reaches the top quintile; stranded capacity is flat across the lower half of the income distribution and declines steeply thereafter, making the facade a transitional configuration rather than an attribute of poverty. Third, the indeterminacy poles separate two configurations calling for opposite interventions, and their composition inverts among facade cases: provision-side misalignment predominates across the population by roughly five to two, but 72% of the top decile of  $\Phi$  is service-side — an inversion that survives setting aside the 18% of economies whose divergence is too small to classify — concentrated in the assessed rather than the physically measured component of state capacity. Fourth, the augmentation reorders human development substantially without overturning it (Spearman  $\rho = 0.947$ ; 77% of economies move three or more positions), and that ordering is reproduced under arithmetic and harmonic aggregation and under full-triple neutrosophic aggregation ( $\rho = 0.999$ ).

## 1 Introduction

Since its 1990 launch by the United Nations Development Programme (UNDP), the Human Development Index (HDI) has become the most widely used composite measure of national well-being, translating Sen’s capabilities approach (Sen, 1985; Anand and Sen, 1994) into a tractable

three-dimensional score covering health, education, and income. Its 2010 methodological reform — the shift from arithmetic to geometric mean aggregation to penalise dimensional imbalance — remains a landmark case of composite-indicator practice adapting to conceptual critique (UNDP, 2010; Klugman et al., 2011). Yet the index was designed in the pre-internet era, and its structure has not evolved to reflect the digital foundations through which citizens now access education, healthcare, income, and public services. Ravallion (2011) and Fukuda-Parr (2003) argued more than a decade ago that the HDI would need to accommodate new capability dimensions as they became constitutive of well-being. Digital access is now such a dimension.

Measures that have attempted to capture it, however, treat what states provide and what citizens actually use as if they were the same thing. They are not, and the gap between them is large. In 184 economies assessed here, the median state has built e-government capacity that reaches only four citizens in five; in the most extreme case the ratio is fourteen to one. Prevailing composite measures fold the two quantities into a single compensatory score. The European Commission’s Digital Economy and Society Index aggregates a digital-public-services dimension, built on e-government provision, with connectivity and citizen-use dimensions in a weighted arithmetic mean, so that strength in state provision offsets weakness in citizen reach directly (European Commission, 2022). The International Telecommunication Union’s ICT Development Index likewise combines infrastructure and access indicators with individual-use indicators in a single composite (International Telecommunication Union, 2024). Under any such aggregation, a gap between what a state provides and what its citizens can reach is averaged into invisibility. We call this masked disjunction the *government facade*.

The scholarly literature has diagnosed the underlying problem without producing a measure of it. One strand documents the digital divide as a stratified inequality with structural rather than transient character (Van Dijk, 2005; Norris, 2001), showing that access, skills, and outcomes form nested layers of exclusion that market provision alone does not close; policy-facing work in the same vein frames the divide through the capabilities approach, as an unequal distribution of digital opportunity rather than a mere infrastructure gap (ESCAP, 2019). A second strand establishes empirical linkages between technological capability and development outcomes: Daghigh Yazd et al. (2025), in a longitudinal panel of twenty-seven countries, find significant positive effects of high-technology exports and research intensity on the HDI, and argue that any twenty-first-century measure of national capability must incorporate technological reach. A third critiques the standard index for smoothing over heterogeneity that matters for policy (Smolińska-Bryza et al., 2025), while Gulmammadov (2025) argues that the divide’s indeterminate, non-linear character exceeds the modelling capacity of deterministic frameworks. Each strand acknowledges that the digital dimension matters; none yields a framework that both augments the HDI and represents the internal structure of digital capability. To our knowledge, no digital augmentation of the HDI proposed to date holds provision-side and use-side indicators analytically apart.

Composite-indicator methodology nonetheless supplies the ingredients. The Multidimensional Poverty Index (Alkire and Foster, 2011) demonstrates that development measurement can integrate multiple deprivation dimensions with rigorous aggregation properties. Fuzzy-logic extensions have gained traction in development monitoring: D’Adamo et al. (2025) show that fuzzy

aggregation of Sustainable Development Goal indicators surfaces North–South inequalities that crisp composites smooth over, and Seidu et al. (2025) integrate the Gini coefficient with fuzzy synthetic evaluation to model non-linear dispersion in infrastructure benefits. Closest to the present concern, Mahdiloo et al. (2023) treat digital-inclusion sub-indices as non-compensatory, on the reasoning that strength in one sub-index should not offset weakness in another, and report partial indices so that localised weakness remains visible. That design prevents offsetting *across* sub-indices; the facade is a disjunction *within* a single dimension, between what a state has built and who can reach it, and no weighting scheme across sub-indices surfaces it. What is required is a decomposition of the dimension itself. Neutrosophic set theory (Smarandache, 1998, 2005) provides one, representing a quantity as an ordered triple of truth, indeterminacy, and falsity components that vary independently. The framework below takes from this apparatus the decomposition and the operators defined on the triple, not a fuzzification step: no membership functions or fuzzy aggregation operators are used (Section 2.4).

Applied to e-government, the triple specifies all three components, each sourced on its own terms. Truth is the published value of the United Nations E-Government Development Index (EGDI): verified state capacity, a documented and cross-country-comparable measure of what a government has built. Falsity is the non-using share of the population: verified non-reach, read throughout as an upper bound, since intermediated access through agents, family members, and assisted-access centres allows some non-users to obtain services indirectly. Indeterminacy is the *signed* divergence between the EGDI’s Online Services and Telecommunications Infrastructure sub-indices, specified as a bipolar component in the sense of Deli et al. (2015), so that its two poles record opposite misalignments rather than a single magnitude. Truth and falsity combine conjunctively to give the magnitude of the gap between provision and reach; the poles of the indeterminacy component give its configuration, and therefore which of two opposite interventions a state requires.

**Contributions.** The paper makes four.

*It supplies a validated measure of a phenomenon previously described but not quantified.* Stranded capacity  $\Phi$  — the absolute share of nominal e-government provision lying beyond citizen reach — and the facade ratio  $R_{\text{facade}}$  render the facade measurable from two published series. More importantly, the measure is tested against evidence external to the indices that construct it. Conditional on state capacity and income level,  $\Phi$  predicts lower citizen uptake of digital government services in demand-side household survey data ( $-0.462$ ,  $p = 2.3 \times 10^{-5}$ ,  $N = 145$ ), and does not predict digital payment activity in general. The measure captures state-mediated digital uptake specifically, which is what it claims to capture and is not what a proxy for connectivity would do.

*It establishes the empirical structure of the phenomenon at full population.* Across 184 economies, facade intensity is concentrated in sub-Saharan Africa and Southern Asia, and the concentration survives an income control. No economy of the Global North reaches the top quintile. Stranded capacity is flat across the lower half of the income distribution and declines steeply thereafter, so the facade is a transitional configuration rather than an attribute of poverty: the poorest economies have not built the capacity to strand.

*It identifies a typology with policy content, and a compositional asymmetry that invites expla-*

*nation*. The bipolar indeterminacy component separates service-side facades, where provision has been built ahead of the network carrying it, from provision-side facades, where the reverse holds. The two call for opposite interventions. Across the population, provision-side misalignment predominates by roughly five to two; among the top decile of stranded capacity the ratio inverts, to 72% service-side, and the reversal is unaffected by setting aside the economies whose divergence is too small to classify. Since the service sub-index is scored from an assessment of what governments report while the infrastructure sub-index is computed from telecommunications statistics, the economies furthest from translating capacity into citizen benefit are disproportionately those whose capacity sits in its assessed component.

*Methodologically, it supplies a neutrosophic decomposition for composite indicators whose components are drawn from different datasets*, together with the operators defined on that triple, an exact normalisation-invariance result (Section 2.3), and a demonstration that a component may be ordinaly inert in aggregation while carrying substantial diagnostic content through its sign. The architecture is available to any indicator facing the same sourcing problem.

The framework is applied at three coverage tiers, set by indicator availability rather than selection: the facade analysis on 184 economies, a three-input digital sub-index on 182, and the full HDI-N composite on 158 (Section 2.2). Every substantive design choice is exposed to a disclosed sensitivity protocol.

The remainder of the paper is organised as follows. Section 2 describes the data sources and measurement framework, including the neutrosophic decomposition, the geometric-mean aggregation, and the disparity operators. Section 3 presents the facade decomposition and typology, the digital sub-index, and the HDI-N composite. Section 4 reports the robustness protocol. Section 5 discusses the findings, sets out the external validation, and acknowledges limitations. Section 6 concludes.

## 2 Methods

This section proceeds as follows. Section 2.1 sets out the framework as a whole. Section 2.2 describes the country panel, indicator selection, reference year, and missing-value handling, and discloses two data-provenance issues that affect interpretation. Section 2.3 states the normalisation rules used to project each raw indicator onto the unit interval. Section 2.4 presents the neutrosophic decomposition of e-government capacity and defines the adjusted score used in downstream aggregation. Section 2.5 defines the digital sub-index. Section 2.6 defines the HDI-N composite. Section 2.7 defines the disparity analysis and its two operators. Section 2.8 specifies the robustness protocol used to evaluate the framework’s sensitivity to modelling choices.

### 2.1 Framework overview

The framework extends the standard HDI by adding a fourth dimension that captures digital capability, and by decomposing the inputs to that dimension neutrosophically so that state provision and citizen reach are held analytically separate. The augmented composite, which we term HDI-N (*neutrosophic*), is a geometric mean of four dimensional sub-indices: health, education, income, and digital. Retaining the geometric aggregator preserves the imbalance-

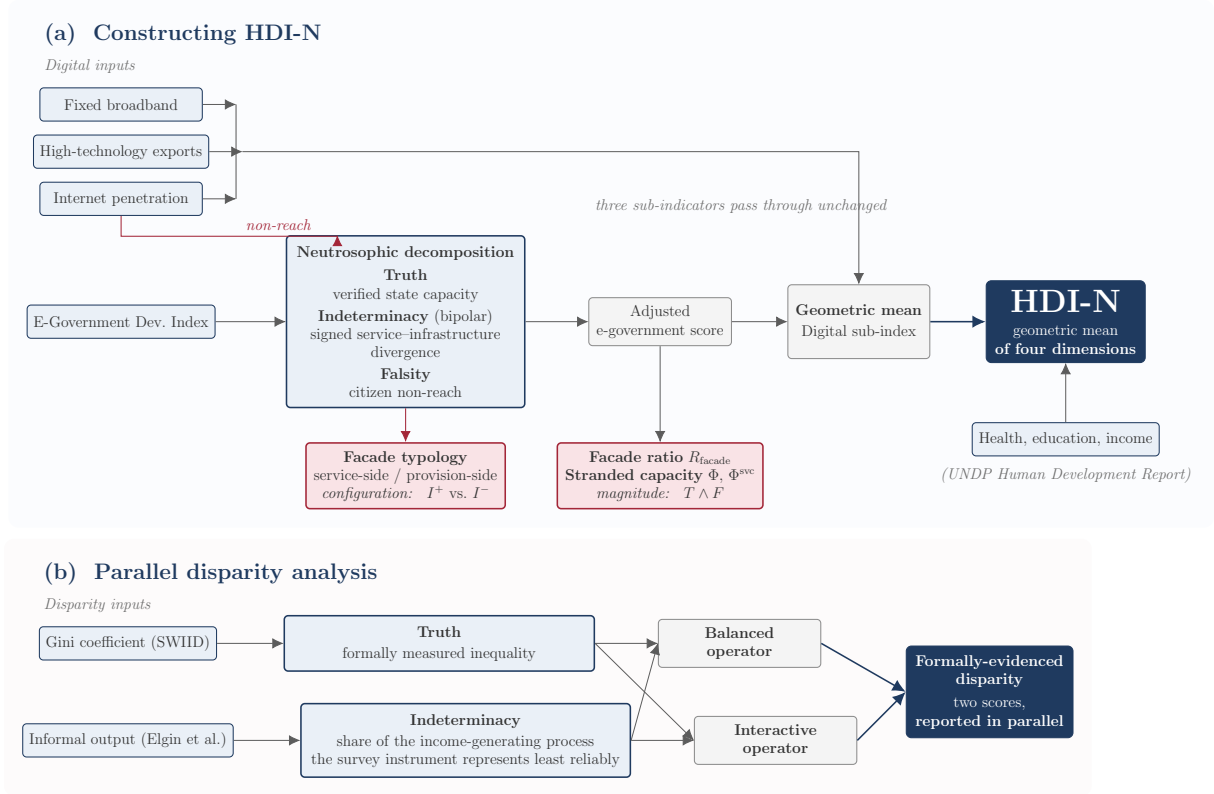


Figure 1: Schematic of the framework. **(a)** Four normalised digital indicators enter the construction, but only the E-Government Development Index is decomposed; fixed broadband, high-technology exports, and internet penetration pass through unchanged to the aggregation stage. The decomposition draws its truth component from the published index value, its falsity component from the share of the population not using the internet — so that internet penetration enters twice, once as a sub-indicator in its own right and once, along the red path, as the source of non-reach — and its indeterminacy component from the signed divergence between the index’s own service and infrastructure sub-components. That double entry is deliberate and is tested directly in the robustness protocol (Section 2.8). The three components are put to two uses. Truth and falsity combine conjunctively to give the *magnitude* quantities: an adjusted e-government score, from which the facade ratio and stranded capacity are read off. The indeterminacy component is bipolar, and its two poles give the *configuration*: whether services have been built ahead of the network carrying them or the reverse. Indeterminacy does not enter the main-text aggregation, a parsimony choice whose ordinal consequences are established in Section 4.4. A geometric mean of the adjusted score and the three remaining sub-indicators forms the digital sub-index; a second geometric mean combines it with the standard health, education, and income dimensions to yield HDI-N. Internet use and the EGDI are normalised against natural upper bounds, fixed broadband and high-technology exports against the maximum over the estimation population, a choice whose consequences for country ordering are bounded exactly in Section 2.3. **(b)** The disparity analysis applies the same separation to inequality, and is reported in the supplementary materials. The SWIID Gini coefficient supplies the truth component and modelled informal output the indeterminacy component, the latter indexing how reliably the survey instrument represents the income-generating process rather than any claim that part of the economy goes unmeasured. Both components feed both operators, which are reported in parallel rather than combined. Definitional detail for each step, including the algebraic form of every operator shown here, is given in Sections 2.3–2.7.

penalising property that motivated the UNDP’s own 2010 methodology reform (UNDP, 2010; Klugman et al., 2011): a country cannot compensate for weak performance in one dimension by exceptional performance in another. The novelty of the framework lies in the construction of the digital dimension itself, which decomposes the UN EGDI into a truth component (measured state capacity), a falsity component (citizen non-penetration), and a bipolar indeterminacy component (the signed divergence between the index’s own service and infrastructure sub-components), following the neutrosophic set-theoretic conventions of Smarandache (1998, 2005) and the bipolar extension of Deli et al. (2015).

The three components are put to two analytical uses, and the distinction runs through the paper. Truth and falsity combine conjunctively to give the *magnitude* of the gap between provision and reach; the poles of the indeterminacy component give its *configuration*, and so identify which of two opposite interventions a given state requires. Because the facade analysis needs only the EGDI and internet penetration, it is estimated on a substantially larger population than the composite (Section 2.2). A parallel disparity analysis applies the same separation to inequality, pairing the formal Gini coefficient with modelled informal output as an indeterminacy component; it demonstrates that the architecture transfers beyond the digital case and is reported in the supplementary materials.

## 2.2 Population, coverage, and missing-value handling

The analysis covers every economy for which the required indicators are published, rather than a selected panel. Because the three analytical layers draw on different indicators, and those indicators differ in coverage, the number of economies declines as the analysis deepens. We set out that structure first, since every subsequent result is reported against one of three populations.

**Coverage cascade.** Table 1 states the three tiers. Tier I supports the facade analysis and requires only the EGDI and internet penetration, both published for essentially all UN member states. Tier II adds fixed-broadband subscriptions and the standard HDI dimensional indices, and supports a three-input digital sub-index. Tier III adds high-technology exports and supports the four-input composite reported in Section 2.6. Each tier admits a construct the tier above cannot represent and pays a coverage price for it; the price is set by indicator availability, not by selection.

Table 1: Coverage cascade. Each tier adds an indicator and the economies for which that indicator is unavailable leave the population. Losses are attributed to the single indicator responsible.

| Tier | Analysis             | $N$ | Additional requirement            |
|------|----------------------|-----|-----------------------------------|
| I    | Facade decomposition | 184 | EGDI 2024, internet penetration   |
| II   | Digital capability   | 182 | + fixed broadband, HDI dimensions |
| III  | Full composite       | 158 | + high-technology exports         |

**Tier I.** The UN publishes an EGDI for all 193 member states. Nine have no ITU penetration value within the 2020–2023 window and are recorded as missing rather than imputed: the Democratic People’s Republic of Korea, Samoa, South Sudan, Sudan, the Syrian Arab Republic,

Turkmenistan, Tuvalu, Vanuatu, and Yemen. Several are economies in which a facade would be plausible, so their absence is a limitation of source coverage rather than a neutral exclusion. A further 36 economies appear in the ITU series but not in the EGDI, being territories or non-member states; they are outside the frame of an analysis of state e-government provision and are listed in the supplementary materials. Tier I accordingly comprises 184 economies.

**Tier II and Tier III.** Two economies leave at Tier II: Chad, which reports no fixed-broadband value, and Monaco, for which the UNDP publishes no dimensional indices. Twenty-four leave at Tier III, every one of them because high-technology exports are unavailable or reported as zero in the 2020–2023 window. That series is the binding constraint on the composite and the only one; broadband and HDI coverage are near-complete. The exclusions are not random: they fall disproportionately on economies with a narrow manufacturing base, and they include Bangladesh, which ranks sixth on stranded capacity at Tier I. The three-input variant of Section 4.5 is defined for these economies and is reported for that reason.

**Reference year and carry-forward rule.** The reference year is 2023 for all digital and inequality indicators, with one exception: the EGDI is taken from the 2024 edition, reflecting the biennial release cycle of the UN survey. Where an indicator is missing for 2023, the most recent value within 2020–2023 is substituted and the year recorded at cell level; beyond that window the indicator is recorded as missing and no imputation is performed. Six economies use a carried-forward penetration value: Eritrea, Myanmar, and Pakistan from 2020, Timor-Leste and Tunisia from 2021, and Botswana from 2022. Pakistan’s is the consequential case and is treated separately in Section 4.6. All series were retrieved in 2026, with retrieval dates recorded at file level and at cell level wherever a carry-forward was applied.

**Indicators and sources.** The following series are used.

- **Individual internet use (% of population).** International Telecommunication Union (ITU) DataHub, series `i99H`.
- **Fixed-broadband subscriptions (per 100 inhabitants).** ITU DataHub, series `i992b`. A second DataHub series (`i4213tfbb`) carries the same unit label but reports raw subscription counts; only `i992b` is used.
- **High-technology exports (% of manufactured exports).** World Bank World Development Indicators (WDI), series `TX.VAL.TECH.MF.ZS`.
- **EGDI and its sub-indices** (Online Services, Telecommunications Infrastructure, Human Capital). UN Department of Economic and Social Affairs, *E-Government Survey 2024*.
- **Standard HDI sub-indices** ( $I_{\text{health}}$ ,  $I_{\text{education}}$ ,  $I_{\text{income}}$ ). UNDP *Human Development Report 2023–2024*, dimensional index values on the UNDP goalpost scale.
- **Gini coefficient, disposable income.** SWIID version 9.x (Solt, 2020), column `gini_disp`, with standard errors from `gini_disp_se`. Used in the disparity analysis only (supplementary materials).

- **Informal output (% of official GDP).** World Bank Informal Economy Database (Elgin et al., 2021), MIMIC estimates, with the dynamic-general-equilibrium estimates from the same source as a robustness check. Used in the disparity analysis only.
- **Citizen uptake of digital financial and government services.** Global Findex Database 2025 (Demirgüç-Kunt et al., 2025), country-level adult population shares for three indicators: received government payments into an account; used a mobile phone or the internet to access an account; made or received any digital payment. These are demand-side household-survey aggregates and enter no index constructed here. They are used only in the external validation specified in Section 2.8 and reported in Section 3.6.

**Provenance disclosures.** Six issues warrant statement.

*The high-technology indicator distorts at both extremes.* As a share of manufactured exports it is inflated where an economy serves as an assembly and re-export node, and where the manufactured-export denominator is narrow. The Philippines illustrates the first case: its 2023 WDI value of 63.98% sets the panel maximum, while OECD Trade in Value Added statistics put domestic value added in Philippine high-technology exports near 47% for 2022 (OECD TiVA, 2024). Iceland illustrates the second: a denominator dominated by aluminium turns a modest absolute volume of pharmaceutical and medical-device exports into a large share. We retain the raw WDI measure for cross-panel consistency, TiVA coverage being incomplete, and treat both distortions as documented limitations in Section 5.5.

*Pakistan’s penetration vintage.* The ITU DataHub reports no penetration value for Pakistan in 2021–2023, a gap re-verified against the source on 27 July 2026, while the 2024 release reports 57.25% against the 2020 carry-forward of 18.94%. A discontinuity of that size reflects revised measurement rather than organic growth. The audited carry-forward is used throughout, and Section 4.6 treats the two figures as interval endpoints and reports which claims are sensitive to the choice.

*Informal-output estimates are modelled, not measured.* The MIMIC and dynamic-general-equilibrium series of Elgin et al. (2021) are model-based estimates whose identifying assumptions have been contested in the literature. They are used only in the supplementary disparity analysis, both are reported, and no claim in the main text rests on either. Their series terminates in 2020 against a 2023 reference year; movement over 2018–2020 averages 1.1 percentage points under MIMIC and 0.6 under the dynamic-general-equilibrium estimates, which bounds the error the mismatch can introduce without licensing extrapolation to 2023.

*SWIID and WDI Gini values are not directly comparable.* SWIID applies cross-country comparability adjustments and reweights for typical underreporting in household-survey inequality data. Its values accordingly diverge from WDI Gini values in the expected direction: slightly lower for OECD economies with strong survey infrastructure, notably higher for developing economies where informal-sector participation and top-tail underreporting are structurally larger. Values reported here are SWIID-comparable and should not be cross-referenced against WDI-based studies without adjustment.

*Three economies report saturated penetration.* The ITU records individual internet use at 100.0% for Bahrain, Saudi Arabia, and the United Arab Emirates. Their falsity component

is therefore exactly zero, as is their stranded capacity, and their facade ratio is exactly unity. Under the upper-bound reading of  $F_{\text{egov}}$  adopted in Section 2.4 this is the substantively correct value and not a missing one — no capacity is stranded where no one is unreached — and the three economies are retained throughout the level-based analyses. It is nonetheless a rounding artefact of the source as well as a substantive zero, and it places them outside any analysis conducted on a logarithmic scale; where that arises the exclusion is stated at the point of use.

*Iceland's Gini value is carried forward from 2019.* SWIID publishes no `gini_disp` value for Iceland within the 2020–2023 window, and the 2019 value (25.3, standard error 0.97) is carried forward. This is the only carry-forward in the inequality series, and its four-year mismatch against the 2023 reference year spans the COVID-19 period, over which the distributional position of high-income economies moved by more than the series' usual year-on-year variation. Iceland's placement in the supplementary disparity analysis should be read with that risk in view.

### 2.3 Normalisation of sub-indicators

Let  $\mathcal{P}$  denote the country panel, and write the raw digital inputs as  $U$  for individual internet use (percentage of population),  $B$  for fixed-broadband subscriptions per 100 inhabitants, and  $H$  for high-technology exports as a percentage of manufactured exports. Sources for each are given in Section 2.2. A country index is suppressed throughout except where a panel-wide extremum is taken. Each digital sub-indicator is projected onto the unit interval  $[0, 1]$  before aggregation.

Individual internet use has a natural upper bound and is rescaled against it:

$$\mathbf{I}_{\text{penetration}} = \frac{U}{100}. \quad (1)$$

The EGDI is published as a 0–1 composite and therefore requires no rescaling; it enters the framework through the decomposition of Section 2.4 rather than through this step.

Fixed-broadband subscriptions and high-technology exports have no theoretical upper bound and are normalised against the panel maximum:

$$\mathbf{I}_{\text{broadband}} = \frac{B}{\max_{c \in \mathcal{P}} B_c}, \quad (2)$$

$$\mathbf{I}_{\text{hitech}} = \frac{H}{\max_{c \in \mathcal{P}} H_c}. \quad (3)$$

Panel-relative normalisation is a recognised limitation of composite-indicator practice, in that the sub-index values it produces are conditional on panel composition (OECD/JRC, 2008). Its consequences here are bounded exactly rather than approximately. Equations 2 and 3 divide by a quantity that is fixed across countries, and the aggregation used throughout is multiplicative (Sections 2.5 and 2.6), so a change of divisor multiplies every country's composite by the same factor.<sup>1</sup> The *within-panel ordering* is accordingly invariant to the choice of divisor: substi-

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<sup>1</sup>Write the divisors as  $c_B = \max_{c \in \mathcal{P}} B_c$  and  $c_H = \max_{c \in \mathcal{P}} H_c$ , both common to every country. Each aggregation step is a product of its inputs raised to fixed positive powers, so  $B$  and  $H$  reach the composite only as  $B/c_B$  and  $H/c_H$  raised to a common exponent, and the divisors contribute the country-invariant factor  $(c_B c_H)^{-1/16}$ . Replacing either divisor therefore rescales all composites by a common positive constant, leaving their order unchanged.

tuting any fixed goalpost, or the maximum of a reduced panel, leaves the ranking of a given set of countries unchanged. What panel composition does affect is the *level* of the composite and its comparability across panels — HDI-N scores computed on different panels are not directly comparable — and the panel-selection question is treated as a limitation in Section 5.5. The invariance does not extend to the min–max normalisation used for the alternative disparity operationalisation of Section 2.7, where subtracting the panel minimum breaks the scale-invariance argument. We adopt panel-relative normalisation for its interpretability under an unbounded raw scale and report the corresponding sensitivity checks in Section 2.8.

One point of implementation follows and should be stated, since the invariance result licenses it. The extremum in Equations 2 and 3 is taken over the population on which the specification concerned is estimated, not over a single fixed panel: Tier III (158 economies) for the four-input composite of the main text, where Andorra sets  $c_B$  and the Philippines  $c_H$ ; Tier II (182) for the three-input variant of Section 4.5, where only  $c_B$  enters. The two variants accordingly report sub-index values on different scales, and their *levels* should not be compared across tables even though, by the argument above, their orderings are unaffected by the difference.

The standard HDI sub-indices  $\mathbf{I}_{\text{health}}$ ,  $\mathbf{I}_{\text{education}}$ , and  $\mathbf{I}_{\text{income}}$  are taken from the 2023–2024 Human Development Report and are already normalised on the UNDP goalpost scale. The income sub-index applies a natural-logarithm transformation of gross national income (GNI) per capita in purchasing-power-parity (PPP) terms, capped at 75,000 dollars, following UNDP convention since 2010. We adopt these sub-indices without modification.

## 2.4 Neutrosophic decomposition of e-government capacity

Neutrosophic set theory (Smarandache, 1998, 2005) represents membership as an ordered triple  $(T, I, F)$ :  $T$  is the degree of verified truth, evidence for membership;  $F$  the degree of verified falsity, evidence against; and  $I$  the degree of indeterminacy, evidence that neither confirms nor negates. Unlike fuzzy membership, which is a single scalar in  $[0, 1]$ , the three components vary independently and are not required to sum to unity. That independence suits composite-indicator work in which the confirming, negating, and uncertain components of a phenomenon are drawn from different sources.

One notational convention is needed before proceeding, since the letter  $I$  carries two meanings in this paper. Plain italic  $I$  always denotes a neutrosophic indeterminacy component, including the e-government indeterminacy defined below. Dimensional and sub-indicator indices of the composite are set in bold throughout —  $\mathbf{I}_{\text{health}}$ ,  $\mathbf{I}_{\text{digital}}$ ,  $\mathbf{I}_{\text{penetration}}$ , and so on — so that no occurrence of  $I$  is ambiguous.

The framework adopts the neutrosophic *decomposition*, the separation of a measured quantity into separately sourced components, together with the operations defined on the resulting triple. It employs no fuzzy membership functions, no fuzzy aggregation operators, and no fuzzification step; the normalisation of Section 2.3 is a projection onto the unit interval, not a membership assignment. The fuzzy-indicator literature cited in Section 1 is precedent for representing partially measurable phenomena in composite indicators, not a description of the present method.

**The roles of the three components.** Deneutrosophication — the collapse of a triple onto a single scalar — is one operation available on  $(T, I, F)$ , not the definition of using it. Because the three components vary independently, they can be put to different analytical uses within a single framework, and this one uses all three in two distinct ways. Truth and falsity combine conjunctively to yield a *magnitude*: how much capacity lies beyond citizen reach (Equations 13 and 14). Indeterminacy yields a *configuration*: which of two opposite misalignments a given state exhibits, and therefore which intervention it calls for (Section 5.1). The magnitude quantities deliberately exclude indeterminacy, for reasons given below; the configuration is defined on it alone. Separating the two is what allows a component that carries little ordinal information about the level of capacity to carry a great deal of policy information about its structure.

For the e-government dimension, the triple is operationalised as follows.

**Truth.** The truth component, denoted  $T_{\text{egov}}$ , is the EGDI value published by the UN for 2024:

$$T_{\text{egov}} = \text{EGDI}_{2024} \in [0, 1]. \quad (4)$$

It represents *verified state capacity*: a documented, cross-country-comparable measure of what a government has built in online services, telecommunications infrastructure, and the human capital required to operate an e-government system. The EGDI is used throughout for the composite, since a dimension of human development requires the broad capacity measure rather than any one of its parts.

For the facade diagnostic we additionally report a second truth component, the Online Services Index alone:

$$T_{\text{egov}}^{\text{svc}} = \text{OSI} \in [0, 1]. \quad (5)$$

This is a narrower construct — deployed service provision, rather than provision together with the infrastructure and human capital supporting it — and it is not a correction of Equation 4 but a measurement of a different quantity. Its role is set out under *Stranded capacity* below.

**Falsity.** The falsity component, denoted  $F_{\text{egov}}$ , is the share of the population that does not use the internet:

$$F_{\text{egov}} = 1 - \mathbf{I}_{\text{penetration}}. \quad (6)$$

It represents *verified non-reach*: the population for whom the state’s digital provision is, definitionally, of no direct use. This is a deliberate upper bound. Intermediated access — through agents, family members, or assisted-access service centres, models well documented in South Asia — allows some non-users to obtain e-government services indirectly, so true non-reach is weakly smaller than  $F_{\text{egov}}$ , and the facade quantities defined below inherit that direction as upper bounds on the gap between provision and effective benefit.

At the upper limit of the source scale the component vanishes: three economies are recorded at 100% penetration and therefore carry  $F_{\text{egov}} = 0$  exactly (Section 2.2). Under the upper-bound reading this is the correct value rather than a missing one, and it is retained; its only consequence is that these economies leave any analysis performed in logarithms.

**Indeterminacy, as a bipolar component.** The EGDI is itself constructed from three sub-indices: the Online Services Index (OSI), the Telecommunications Infrastructure Index (TII), and the Human Capital Index. Their signed difference,

$$D_{\text{egov}} = \text{OSI} - \text{TII} \in [-1, 1], \quad (7)$$

records a misalignment between the service capacity a state reports and the infrastructure carrying it. Divergence neither confirms nor negates the capacity the index reports: a state may have built services its network cannot deliver, or a network on which few services yet run, and the aggregate index value is compatible with either. That is indeterminacy in the sense of Smarandache (1998) — evidence that qualifies the truth component without negating it, which is the role  $F_{\text{egov}}$  already fills.

The two configurations are not merely different values of one quantity; they are opposites, and they call for opposite interventions. The indeterminacy component is therefore specified as *bipolar* in the sense of Deli et al. (2015), who extend the neutrosophic triple so that each component carries a positive degree in  $[0, 1]$ , denoting membership in a property, and a negative degree in  $[-1, 0]$ , denoting membership in its anti-property. We set

$$I_{\text{egov}}^+ = \max(D_{\text{egov}}, 0) \in [0, 1], \quad I_{\text{egov}}^- = \min(D_{\text{egov}}, 0) \in [-1, 0], \quad (8)$$

so that the positive pole records service provision built ahead of the infrastructure to deliver it, and the negative pole the inverse. The magnitude of the component,

$$I_{\text{egov}} = |I_{\text{egov}}^+ + I_{\text{egov}}^-| = |D_{\text{egov}}|, \quad (9)$$

is what enters the full-triple aggregation of Section 4.4; the poles carry the configuration, and support the typology developed in Section 5.1.

Three properties of this specification should be stated rather than left to be discovered. First,  $D_{\text{egov}}$  is a single scalar, so *at most* one pole is non-zero for any economy: the poles are mutually exclusive by construction, where a general bipolar set admits simultaneous membership in a property and its anti-property. The excluded case  $D_{\text{egov}} = 0$ , in which both poles vanish and the economy is aligned, is not empty in practice and is taken up under the third property below. Second, bipolarity is claimed for the indeterminacy component alone. Neither truth nor falsity has a natural anti-property here, so their negative degrees are identically zero and the construction is a partial rather than a complete bipolar neutrosophic set. Both of these are consequences of adopting bipolarity because the phenomenon is bipolar, rather than importing the full apparatus for its own sake.

Third, and least obviously, the sign of  $D_{\text{egov}}$  is informative only where its magnitude exceeds the resolution of the quantities from which it is formed. The OSI and TII are published to three decimal places, and each is itself a composite — the first of an assessment conducted against a questionnaire, the second of reported telecommunications statistics — so a divergence of a few thousandths is not evidence of a configuration but of two noisy composites failing to coincide exactly. Reading such a case as “services built ahead of the network” would assert a policy conclusion that the data cannot carry. We therefore retain the sign classification, which is what

the bipolar construction yields, but pair it throughout with an *aligned band*

$$|D_{\text{egov}}| \leq \delta, \quad \delta = 0.05, \quad (10)$$

within which no configuration is asserted. The threshold is a judgement about measurement resolution rather than an estimate, and Section 3.2 reports the population composition under both readings and establishes that no substantive claim depends on economies inside the band.

**Construct overlap among the components.** One property of this operationalisation must be stated plainly, because it qualifies the claim that the components are separately sourced.

$I_{\text{egov}}$  is computed from the EGDI's own sub-indices. It is therefore not sourced independently of  $T_{\text{egov}}$  and cannot be: it is a *within-index* misalignment diagnostic, and is better read as internal structure in the state-capacity signal than as indeterminacy drawn from outside it. No available data source supplies an externally measured indeterminacy for e-government capacity at cross-country scale; an independent construction would require a measure of actual service uptake alongside the assessed provision measure, which exists for OECD and EU members but not for the economies where the facade is largest (Section 5.6).

A second overlap runs between  $T_{\text{egov}}$  and  $F_{\text{egov}}$ . The TII is itself built in part from ITU internet-user data, so the EGDI contains internet penetration with an effective weight of approximately one twelfth, and  $T_{\text{egov}}$  is accordingly not fully independent of the series supplying  $F_{\text{egov}}$ . The direction of the resulting bias is determinate and conservative for the facade quantities. Economies with low penetration have a depressed TII and therefore a depressed EGDI, so the product  $T_{\text{egov}} \cdot F_{\text{egov}}$  is attenuated precisely where non-reach is largest: the stranded-capacity figures reported below understate the gap in the cases where it matters most. The narrower truth component of Equation 5 removes this overlap entirely, the OSI being an assessed measure of government service provision that draws on no ITU series, and this is one reason both readings are reported.

The two overlaps cannot be eliminated simultaneously by any choice of truth component drawn from the EGDI, since  $T_{\text{egov}}^{\text{svc}}$  removes the first while entering directly into  $D_{\text{egov}}$ . We therefore disclose both, and keep the facade magnitudes free of indeterminacy — so that the paper's headline diagnostic depends on no overlapping pair.

**Adjusted score.** The e-government score entering downstream aggregation, denoted  $T_{\text{egov}}^{\text{adj}}$ , combines truth and falsity multiplicatively:

$$T_{\text{egov}}^{\text{adj}} = T_{\text{egov}} \cdot (1 - F_{\text{egov}}) = \text{EGDI} \cdot \mathbf{I}_{\text{penetration}}. \quad (11)$$

Nominal state capacity is discounted by the share of the population it does not reach: a state with EGDI 0.9 whose digital services are used by 20% of citizens contributes 0.18 to the digital sub-index, not 0.9. The operator is conjunctive and therefore non-compensatory, consistent with the aggregation principle applied throughout Sections 2.5 and 2.6: a high truth component cannot offset the discount.

Equation 11 combines two of the three components. Section 4.4 carries the complete triple

through the pipeline under two deneutrosophication operators of opposite compensatory character, and establishes that the resulting country ordering is reproduced. The two-component form is therefore retained as the parsimonious operator whose ordering the complete triple reproduces, not as a simplification adopted for convenience; the criterion by which that judgement is made is fixed in advance in Section 2.8.

**Facade ratio.** The government-facade ratio, denoted  $R_{\text{facade}}$ , is given by

$$R_{\text{facade}} = \frac{T_{\text{egov}}}{T_{\text{egov}}^{\text{adj}}} = \frac{1}{\mathbf{I}_{\text{penetration}}}. \quad (12)$$

It expresses how much larger nominal state capacity is than its effective reach: a value of 1 indicates parity between provision and use, and values substantially above 1 indicate that documented capacity outstrips citizen access. The reduction to the reciprocal of the penetration rate is derived from the operationalisation of  $F_{\text{egov}}$  rather than assumed, and isolates the relative overstatement of capacity. Two consequences follow. First,  $R_{\text{facade}}$  alone cannot identify a facade case, since it carries no information about the absolute level of capacity being overstated. Second, the ratio is invariant to the choice of truth component: the numerator cancels, so Equations 4 and 5 yield the same value, and only one ratio is reported throughout.

Because the ratio reduces to a function of a single input, it is worth stating plainly what that reduction does and does not license. As a *quantity*,  $R_{\text{facade}}$  carries no information beyond the penetration rate, and we make no claim that it does. What the construction supplies is a *reading*: the same number, arrived at as the quotient of nominal over reach-adjusted capacity, is the factor by which a published capacity figure overstates the capacity actually within citizens' reach. That interpretation is what is new, not the arithmetic, and it is available only once the truth and falsity components have been separated. The ratio is reported for that reason, is never used on its own to identify a facade case, and is not treated anywhere below as independent evidence alongside  $\Phi$ .

**Stranded capacity.** Stranded capacity, denoted  $\Phi$ , is the conjunction of the truth and falsity components:

$$\Phi = T_{\text{egov}} \cdot F_{\text{egov}}. \quad (13)$$

It measures the amount of nominal e-government capacity lying beyond citizen reach. Neutrosophically,  $\Phi$  is the product t-norm conjunction  $T_{\text{egov}} \wedge F_{\text{egov}}$ : the co-presence of verified state capacity and verified non-reach. The parallel quantity built on the narrower truth component of Equation 5,

$$\Phi^{\text{svc}} = T_{\text{egov}}^{\text{svc}} \cdot F_{\text{egov}}, \quad (14)$$

we term *stranded service capacity*: the share of deployed online service provision lying beyond citizen reach. The two are reported in parallel and neither is a correction of the other.  $\Phi$  answers what share of a state's overall digital capacity is unreachable;  $\Phi^{\text{svc}}$  answers what share of its *deployed services* is unreachable, and the two diverge exactly where a state's service and infrastructure profiles diverge — that is, where  $I_{\text{egov}}$  is large. The relationship between the two orderings is reported in Section 3.1.

Neither magnitude incorporates  $I_{\text{egov}}$ . This is deliberate rather than parsimonious. Both are conjunctions of a verified positive with a verified negative, and the interpretation of each — capacity co-present with non-reach — depends on both arguments being verified quantities. Admitting a component that by construction neither confirms nor negates would leave the product without a reading in the terms that make it useful. The indeterminacy component instead does its diagnostic work through the poles of Equation 8, where it distinguishes two configurations that the magnitudes alone cannot separate.

The quantities divide the interpretive labour three ways.  $R_{\text{facade}}$  gives the *relative* factor by which nominal capacity overstates effective reach and is independent of the capacity level.  $\Phi$  and  $\Phi^{\text{svc}}$  give the *magnitude* of capacity stranded, and are large only where substantial capacity and substantial non-reach co-occur. The bipolar component gives the *configuration* of the misalignment. A facade case, in the sense developed in Section 3.1, is an economy extreme on a magnitude and on the ratio together; which configuration it exhibits, and therefore which intervention it calls for, is read from the pole.

## 2.5 The digital sub-index

The digital sub-index, denoted  $\mathbf{I}_{\text{digital}}$ , is the geometric mean of the four normalised digital inputs:

$$\mathbf{I}_{\text{digital}} = \sqrt[4]{\mathbf{I}_{\text{penetration}} \cdot \mathbf{I}_{\text{broadband}} \cdot T_{\text{egov}}^{\text{adj}} \cdot \mathbf{I}_{\text{hitech}}} \quad (15)$$

The four inputs represent complementary rather than substitutable aspects of digital capability: individual internet use captures baseline access; fixed-broadband subscriptions the depth and quality of connectivity; the adjusted e-government score of Equation 11 effective state provision; and high-technology exports the economic translation of digital capability into internationally tradeable output. This fourth input is operationalised as *goods* conversion specifically — high-technology merchandise exports as a share of manufactured exports (Section 2.2), following Daghigh Yazd et al. (2025) — and does not capture digital *services* trade, a scope restriction whose consequences for the OECD conversion findings are taken up in Section 5.4.

The aggregator requires all four inputs to be present and strictly positive: a missing input leaves  $\mathbf{I}_{\text{digital}}$  undefined, and a zero input drives it to zero. Neither case is imputed away. This follows from the framework’s substantive claim that digital capability is genuinely multidimensional, so that a country lacking one dimension is not adequately described by a composite that averages over its absence.

## 2.6 The HDI-N composite

The augmented composite, denoted HDI-N, is the geometric mean of the four dimensional sub-indices, following the aggregation the UNDP adopted in 2010:

$$\text{HDI-N} = \sqrt[4]{\mathbf{I}_{\text{health}} \cdot \mathbf{I}_{\text{education}} \cdot \mathbf{I}_{\text{income}} \cdot \mathbf{I}_{\text{digital}}} \quad (16)$$

The choice of aggregator reflects the argument that motivated the UNDP’s own shift from arithmetic to geometric mean: dimensional imbalance is not correctable through overperformance elsewhere. A country with strong health, education, and income indicators but limited digital

capability sees its HDI-N reduced in proportion to the geometric penalty on imbalance. The same non-compensatory principle has been argued directly in the digital-indicator setting by Mahdiloo et al. (2023), who reject additive aggregation of digital-inclusion sub-indices on the ground that it lets a weak sub-index be offset, or zeroed out entirely, by strength elsewhere; the geometric mean imposes the analogous restriction here through its imbalance penalty rather than through a programming constraint.

This property carries the framework’s substantive interpretation. HDI-N is not a re-weighting of the existing HDI dimensions but an augmented composite in which digital capability is treated as constitutive of contemporary human capability rather than incidental to it.

## 2.7 Disparity analysis

The HDI-N composite measures aggregate national capability without attention to within-country distributional dispersion. We therefore conduct a parallel disparity analysis applying the same neutrosophic separation to the Gini coefficient. Its purpose is to demonstrate that the architecture transfers beyond the digital case: the operators are defined here and the results are reported in the supplementary materials, where the coverage and source sensitivity of the exercise can be set out at the length they require. The construction uses a truth–indeterminacy pair only; no falsity component is defined, for reasons given below. To keep these components distinct from the e-government triple of Section 2.4, they carry the subscript “ineq” throughout.

**Truth.** Let  $G$  denote the SWIID disposable-income Gini coefficient in percentage points (column `gini_disp`), and  $\sigma_G$  its published standard error (column `gini_disp_se`). The truth component, denoted  $T_{\text{ineq}}$ , scales  $G$  to the unit interval:

$$T_{\text{ineq}} = \frac{G}{100}. \quad (17)$$

It represents formally measured inequality: the distributional signal that official household-survey statistics report.

**Indeterminacy.** Let  $E$  denote modelled informal output as a percentage of official GDP (Elgin et al., 2021). The indeterminacy component, denoted  $I_{\text{ineq}}$ , is that share on the unit interval:

$$I_{\text{ineq}} = \frac{E}{100}. \quad (18)$$

The rationale requires precise statement, because the claim concerns measurement *reliability* rather than measurement *coverage*. The household surveys underlying SWIID sample households irrespective of the formality of their income sources; informal workers fall within the sampling frame and report income to enumerators, so the informal economy is not invisible to  $G$ . What varies with informality is how reliably the recorded distribution represents the underlying one. Income generated outside the formal sector is systematically harder to measure: through under-reporting of self-employment and informal earnings relative to wage income, documented even in high-income settings (Hurst et al., 2014); through undercapture of top incomes (Lustig and Vigorito, 2025); and through the widening gap between recorded and actual income concepts

where informal activity is large (Deaton, 2003). The informal-output share accordingly indexes the portion of the income-generating process whose distributional shape the survey instrument represents least reliably.

This is also why the construction defines no falsity component. Falsity in the neutrosophic sense requires a verified negating quantity, as citizen non-reach supplies for e-government (Equation 6). No such quantity exists here: nothing in the data verifies that a determinate portion of the income distribution is misreported. What  $E$  supplies is uncertainty that neither confirms nor negates the measured signal — indeterminacy in the sense of Smarandache (1998) — so the pair  $(T_{\text{ineq}}, I_{\text{ineq}})$  is a complete specification rather than a truncated triple.

The operationalisation is consistent with, rather than duplicative of, the SWIID adjustments disclosed in Section 2.2. SWIID’s harmonisation adjusts estimates toward the typical cross-source and cross-country reporting relationships observed in its calibration data (Solt, 2020), and cannot absorb country-specific, informality-driven measurement failure beyond that typical pattern. Nor does  $\sigma_G$  carry that burden: a standard error quantifies imprecision, whereas informality-driven underreporting is a bias mechanism, and no standard error captures bias.

**Operators.** Two operators are defined on the pair. The balanced operator, denoted  $S_{\text{bal}}$ , is given by

$$S_{\text{bal}} = 0.5 \cdot T_{\text{ineq}} + 0.5 \cdot (1 - I_{\text{ineq}}), \quad (19)$$

and the interactive operator, denoted  $S_{\text{int}}$ , by

$$S_{\text{int}} = T_{\text{ineq}} \cdot (1 - I_{\text{ineq}}). \quad (20)$$

$S_{\text{bal}}$  treats the measured signal and the formal share of the economy as substitutable, weighting each equally.  $S_{\text{int}}$  treats them as multiplicative, discounting the formal signal in proportion to the share of the economy from which reliable measurement can be drawn. The two encode different substantive claims about how signal and reliability combine, and are reported in parallel.

**Interpretation.** The direction of the indeterminacy discount is easily misread, so we state the reading explicitly. Neither operator measures inequality, and neither adjusts  $G$ : both act on the evidential weight the estimate can support. This is what separates them from SWIID’s harmonisation, which acts on the estimate itself. Both operators measure *formally-evidenced disparity*: the magnitude of the inequality signal that formal statistics can be relied upon to support. A high  $S_{\text{int}}$  indicates substantial measured inequality resting on a reliably measured base; a low  $S_{\text{int}}$  indicates either genuinely low measured inequality or — quite differently — a formal estimate whose base is too unreliable to evidence it. The two routes to a low score are substantively opposite and are distinguished only by inspecting  $T_{\text{ineq}}$  and  $I_{\text{ineq}}$  separately, which is why both components are reported alongside the operators. An economy ranked low on  $S_{\text{int}}$  should not be described as equal, nor one ranked high as unequal, without reference to the underlying pair.

**Alternative operationalisations.** Two are reported in the supplementary materials. The first replaces the MIMIC estimates of Equation 18 with the dynamic-general-equilibrium esti-

mates published in the same source, which derive from a different modelling family (Elgin et al., 2021). The second replaces the informal-output proxy entirely with  $\sigma_G$ , min–max normalised across the estimation population:

$$I_{\text{ineq}}^{\text{SE}} = \frac{\sigma_G - \min_c \sigma_{G,c}}{\max_c \sigma_{G,c} - \min_c \sigma_{G,c}}. \quad (21)$$

This captures sampling uncertainty in the Gini estimate rather than indeterminacy in the underlying phenomenon. We adopt the informal-output specification because it parallels the e-government decomposition — both discount a measured truth component by the share of the phenomenon that measurement represents least reliably. Note that the min–max form subtracts the population minimum, so the exact ordering invariance established in Section 2.3 does not extend to this specification. The ordering produced by the disparity analysis is sensitive to which informality source is adopted, and the supplementary materials report that sensitivity in full rather than resolving it.

## 2.8 Robustness protocol

Eight exercises evaluate the framework. The first seven are sensitivity checks: they interrogate the framework’s response to its own modelling choices, and are reported in Section 4. The eighth is of a different kind — a test of whether the central quantity corresponds to anything observed outside the indices that construct it — and because it yields a finding rather than a sensitivity result it is reported with the results, in Section 3.6. All eight are specified here, before the fact.

**Bootstrap rank confidence intervals.** Let  $x$  denote a normalised sub-indicator value and  $\tau$  a relative noise level. For each of 1,000 replicates, every normalised sub-indicator is perturbed additively,

$$\tilde{x} = x + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, (\tau x)^2), \quad (22)$$

with  $\tilde{x}$  clipped to  $[0, 1]$ ; the HDI-N composite is recomputed and country ranks recorded. The 2.5th, 50th, and 97.5th percentiles of each country’s rank distribution are reported as a 95% interval on rank. The analysis is run at  $\tau \in \{0.01, 0.03, 0.05\}$ , corresponding to 1%, 3%, and 5% representative measurement noise, with  $\tau = 0.03$  as the reference level. This order of magnitude is consistent with reported measurement variation in the underlying ITU and WDI series and with noise levels used in comparable composite-indicator sensitivity studies (OECD/JRC, 2008; Aguña and Kovačević, 2010).

Sub-indicators are perturbed independently. This is a disclosed simplification: internet penetration enters the composite twice, directly and inside  $T_{\text{egov}}^{\text{adj}}$  (Equation 11), so a fully specified noise model would perturb the two entries in a correlated fashion. Independent perturbation therefore understates the effective variance attaching to penetration, which is one reason the protocol reports a ladder of noise levels rather than a single value.

**Aggregation-operator sensitivity.** Let  $\mathbf{I}_1, \dots, \mathbf{I}_4$  denote the four dimensional sub-indices of Equation 16. The composite is recomputed under two alternative aggregators: the arithmetic mean  $\frac{1}{4} \sum_j \mathbf{I}_j$ , and the constant-elasticity-of-substitution (CES) operator at substitution

parameter  $\theta = -1$ ,

$$\bar{\mathbf{I}}_{\text{CES}} = \left( \frac{1}{4} \sum_{j=1}^4 \mathbf{I}_j^\theta \right)^{1/\theta}, \quad \theta = -1, \quad (23)$$

which is the harmonic mean. Spearman rank correlations between each pair of orderings (geometric–arithmetic, geometric–harmonic, arithmetic–harmonic) are reported. The interpretive criterion is fixed in advance: correlations near unity would establish that the country ordering is not an artefact of aggregation choice, leaving the geometric mean to be preferred on interpretive grounds alone, whereas correlations substantially below unity would establish that the choice of aggregator is materially consequential and must be defended on stronger grounds than convention.

**HDI-N and standard HDI correlation.** Pearson and Spearman correlations between the HDI-N rank and the standard HDI rank are computed across the countries for which HDI-N is defined. Two thresholds are fixed in advance. A correlation close to unity would indicate that the digital augmentation does not substantively reorder countries; a correlation below approximately 0.8 would indicate that it overrides rather than extends the underlying HDI structure. An intermediate value would establish that the augmentation does analytical work while preserving the interpretability of the HDI tradition.

**Disparity ranking under source-published uncertainty.** The bootstrap is re-run on the  $S_{\text{int}}$  ordering (Equation 20) with  $\sigma_G$ , the SWIID published standard error of the Gini estimate, used as the noise scale on the inequality input in place of a flat relative level  $\tau$ . This substitutes the measurement uncertainty the source itself reports for an assumed one.

**Full-triple neutrosophic aggregation.** The main-text adjusted score of Equation 11 combines the truth and falsity components only, the indeterminacy component doing its work through the poles of Equation 8 rather than through the aggregation (Section 2.4). Whether that division of labour costs anything ordinarily is an empirical question, and this check settles it by carrying the complete triple  $(T_{\text{egov}}, I_{\text{egov}}, F_{\text{egov}})$  through the entire pipeline.

Two deneutrosophication operators are used, chosen because they differ in functional form rather than merely in parameterisation. The first is the multiplicative extension

$$T_{\text{egov}}^{\text{full}} = T_{\text{egov}} \cdot (1 - F_{\text{egov}}) \cdot (1 - I_{\text{egov}}), \quad (24)$$

which completes the main-text operator on its own logic: state capacity is discounted first by the share of the population it does not reach and then by the internal misalignment between service provision and the infrastructure carrying it. Being conjunctive, it is non-compensatory, so a high truth component cannot offset either discount. The second is the standard single-valued neutrosophic score function

$$s_{\text{egov}} = \frac{2 + T_{\text{egov}} - I_{\text{egov}} - F_{\text{egov}}}{3}, \quad (25)$$

established in the neutrosophic decision-making literature (Şahin, 2014; Ye, 2014), which maps

the triple onto  $[0, 1]$ , attaining unity at  $(1, 0, 0)$  and zero at  $(0, 1, 1)$ . Unlike Equation 24, it is additive and therefore compensatory: a higher truth component can offset a higher indeterminacy or falsity component one for one. The two operators consequently embody opposite assumptions about how the components combine, which is what makes them a usable pair for a robustness check rather than a redundant one.

Each operator is substituted for  $T_{\text{egov}}^{\text{adj}}$  at the same point in the pipeline, with Equations 15 and 16 left unchanged, so that any difference in the resulting ordering is attributable to the deneutrosophication step alone. The recomputed HDI-N scores and ranks are compared with the main-text ordering by Spearman rank correlation and by country-level rank displacement. The criterion is stated in advance: if either full-triple ordering departed materially from the main-text ordering, the two-component operator would not be defensible and  $I_{\text{egov}}$  would have to enter the main specification; agreement would establish that the operator is ordinally inert and adopted for parsimony rather than convenience. Where displacement does occur, we additionally report it against  $I_{\text{egov}}$ , since a two-component operator is defensible only if the economies it treats differently are those whose indeterminacy is largest.

**Sensitivity to the composition of the digital sub-index.** The digital sub-index (Equation 15) raises two questions. One input, high-technology exports, measures the economic conversion of digital capability into tradeable output rather than citizen or state digital capacity directly; and internet penetration enters twice, once directly and once inside  $T_{\text{egov}}^{\text{adj}}$ . Both are addressed by a  $2 \times 2$  grid crossing the composition of the sub-index (four inputs; three inputs, omitting high-technology exports) with the treatment of e-government capacity (the adjusted score  $T_{\text{egov}}^{\text{adj}}$ ; the raw EGDI, which removes the second entry of penetration). The four orderings are compared with one another and with the standard HDI ranking. The three-input variants are additionally defined for Bangladesh, whose exclusion from the main composite follows solely from the missing high-technology export series (Section 2.2).

**Sensitivity of the facade characterisation to Pakistan’s penetration vintage.** Pakistan’s penetration input is a 2020 carry-forward (Section 2.2). A seventh check recomputes the facade quantities of Equations 12 and 13, the facade characterisation of Section 3.1, and the HDI-N composite with the 2024 release value adopted as a nowcast, treating the two figures as endpoints of an interval and reporting which claims are sensitive to the choice between them.

**External validation against demand-side survey data.** The seven checks above establish that the framework is stable under its own assumptions. None of them can establish that  $\Phi$  measures anything: a quantity constructed as the product of two published indices can be internally consistent, robust to every perturbation, and correspond to nothing observed. The eighth exercise tests exactly that, against the Global Findex Database 2025 (Demirgüç-Kunt et al., 2025), a demand-side household survey that records what citizens do rather than what states are assessed to provide and which enters no index constructed here.

The specification is fixed in advance. Three Findex country aggregates serve as dependent variables: the share of adults who received government payments into an account (the *state-mediated* indicator), the share who used a mobile phone or the internet to access an account,

and the share who made or received any digital payment (the *broad* indicator). Each is regressed on the EGDI and the UNDP income sub-index with heteroskedasticity-robust standard errors, and  $\Phi$  is then added. The exercise is repeated with  $\Phi^{\text{svc}}$  in place of  $\Phi$  and on the Global South subsample, where the range of  $\Phi$  is widest.

Two criteria are fixed before the estimates are seen, and the second is the demanding one.

*Discriminant criterion.*  $\Phi$  should carry a negative and significant coefficient on the state-mediated indicator and no coefficient on the broad one. A measure that moved both would be behaving as a proxy for connectivity or for development rather than for the gap between provision and reach, since both of those would depress digital activity of every kind.

*Incremental criterion.*  $\Phi$  must outperform its own falsity component. Because  $\Phi = T_{\text{egov}} \cdot F_{\text{egov}}$  and  $T_{\text{egov}}$  is already among the controls, a specification that omits  $F_{\text{egov}}$  cannot distinguish the conjunction from non-reach entered alone — and non-reach carries the larger part of the conjunction’s cross-country variance (Section 3.1). We therefore enter the two together, regressing each dependent variable on the EGDI, the income sub-index,  $F_{\text{egov}}$ , and  $\Phi$  in a common specification. The conjunction earns its place only if it survives the presence of the component whose variance it inherits. Should  $F_{\text{egov}}$  absorb it, the honest conclusion is that stranded capacity is non-reach weighted by a nearly constant factor and should be reported as such; that criterion is recorded here so that the reading of the result is not settled after it is known.

### 3 Results

Results are reported by tier. Section 3.1 presents the facade decomposition on the full Tier I population of 184 economies, and Section 3.2 the facade typology derived from the bipolar indeterminacy component on the same population. Section 3.3 describes the digital indicators and Section 3.4 the digital sub-index. Section 3.5 reports the HDI-N composite on the Tier III population of 158 economies and its rank reordering relative to the standard HDI. Section 3.6 reports the external validation of the facade measure against demand-side household survey data, on the 145 economies for which those data are published. The disparity analysis of Section 2.7 is reported in the supplementary materials. Robustness results are reported separately in Section 4.

#### 3.1 The facade decomposition

The facade quantities require only the EGDI and internet penetration and are therefore estimated on the full Tier I population of 184 economies. We report them as continuous distributions rather than as a classification: the two quantities correlate at  $\rho = 0.90$ , so any joint threshold rule is both nearly redundant and sharply sensitive to where the cut is placed, and the substantive claims below are made on the continuous quantities and checked against a ladder of cutoffs in Section 4.7.

**Distribution.** Stranded capacity has a median of 0.134 and a mean of 0.145 across the population, with an interquartile range of 0.071 to 0.199 and a maximum of **0.453** (Zambia). The facade ratio is more skewed: its median is 1.25, but it reaches **13.94** in Burundi, where an EGDI of 0.25 rests on an access base of 7.2%. The contrast between the two extrema is the reason both

quantities are reported.  $R_{\text{facade}}$  is largest where capacity is smallest relative to reach, which is a different phenomenon from capacity being stranded in absolute terms, and it is the latter that a facade denotes.

Figure 2 places every economy in the plane of state capacity against citizen reach, with contours of constant  $\Phi$ . Because  $\Phi = T_{\text{egov}} \cdot F_{\text{egov}}$ , the contours are the level sets of that product, and an economy’s stranded capacity is read as its displacement above the parity line on which capacity and reach coincide. The facade region is the upper-left of the plane: substantial capacity, limited reach. Its lower-right counterpart — reach running ahead of capacity, as in Chad and Somalia — is the inverse pathology, and the framework registers it as a small  $\Phi$  rather than a facade.

**Stranded capacity is not non-reach relabelled.** A conjunction of this form invites an objection, and it is worth identifying which one. That  $\Phi$  restates the EGDI is not it: the two are negatively associated across the population ( $\rho = -0.62$ ), since the economies with the most capacity are also those whose citizens are least often unreached. The objection that bites is the opposite one — that  $\Phi$  restates non-reach. It has a basis.  $F_{\text{egov}}$  carries substantially more cross-country variance than  $T_{\text{egov}}$ , and the two orderings correlate at  $\rho = 0.90$ . (This is necessarily the same figure as for  $R_{\text{facade}}$ , which is a monotone function of the same input.)

Rank correlation is nonetheless the wrong statistic against which to test the claim, because the measure is not used to order the population; it is used to identify its extremes, and it is there that the two quantities part company. The top decile of  $\Phi$  shares only **five of eighteen** members with the top decile of  $F_{\text{egov}}$ , and at the quintile the overlap is 24 of 37. Burundi, Chad, Somalia, Niger, the Central African Republic, Eritrea, Afghanistan, and Mozambique are all top-decile on non-reach, and not one of them is a facade case; Kenya, Bangladesh, India, Sri Lanka, Nepal, Rwanda, Nigeria, and Tanzania are all top-decile on  $\Phi$ , and not one of them is top-decile on non-reach. The two disagree precisely on the set the measure exists to pick out, which is what the construct requires and what the correlation conceals: the conjunction selects economies that have built capacity *and* fail to deliver it, while non-reach alone selects economies that have built nothing to deliver. The same separation is demonstrated against outcome data, rather than against the components, in Section 3.6.

**Regional structure.** Facade intensity is concentrated, and the concentration survives an income control. Mean  $\Phi$  in sub-Saharan Africa is 0.226 across 47 economies, against 0.117 elsewhere; a one-sided Mann–Whitney test rejects equality at  $p = 2.3 \times 10^{-12}$ . Southern Asia has the higher central tendency — a mean of 0.237 — but on nine economies, so it supplies the most intense cases while sub-Saharan Africa supplies the most numerous. Figure 3 shows both facts in one panel: a right-shifted Southern Asian distribution over few economies, and a sub-Saharan African distribution that is marginally lower but far denser and carries four of the five largest values in the population. Figure 4 maps the same quantity geographically, and shows a feature the distributional views cannot: facades of both configurations occur side by side among neighbouring economies at comparable development levels.

Regressing  $\Phi$  on regional indicators and the UNDP income sub-index, with heteroskedasticity-robust standard errors, gives a sub-Saharan African coefficient of +0.037 ( $p = 0.029$ ) and a

Southern Asian coefficient of  $+0.082$  ( $p = 0.059$ ) against an income coefficient of  $-0.289$  ( $p < 0.001$ ), on  $N = 183$  with  $R^2 = 0.504$ . The single economy lost relative to Tier I is Monaco, for which the UNDP publishes no dimensional indices and which therefore drops with the income control. Income dominates, as expected, but the regional effects are not absorbed by it: the sub-Saharan concentration is not simply a restatement of the income gradient.

**The North–South partition describes but does not explain.** No economy of the Global North appears in the top quintile of  $\Phi$ . The Northern maximum is 0.181 (Republic of Moldova) against a Southern maximum of 0.453, Northern mean  $\Phi$  is 0.074 against 0.173, and a one-sided Mann–Whitney test returns  $p = 1.8 \times 10^{-12}$ . The directional claim is therefore about as clean as such claims get, and it is visible in Figure 2 as the absence of Northern markers from the upper-left region and in Figure 4 as a near-neutral band across North America, Europe, and Australasia.

The explanatory claim does not survive the same control. A North–South indicator alone carries a coefficient of  $+0.098$  ( $p < 0.001$ ,  $R^2 = 0.219$ ); adding the income sub-index reduces it to  $+0.013$  ( $p = 0.242$ ) while income itself enters at  $-0.337$  ( $p < 0.001$ ). The partition tracks where facades are but adds nothing once income is accounted for — in contrast to the sub-Saharan indicator, which survives the same test. We therefore retain North–South as a descriptive frame and locate the analytical claim at the level of income and region.

**Facade is not a poverty phenomenon.** The relationship between  $\Phi$  and development level is not monotone across the range (Figure 5). Stranded capacity is flat across the lower half of the income distribution — octile means of 0.222 and 0.232 in the two lowest octiles, with overlapping bootstrap intervals — and declines steeply only thereafter. A quadratic specification returns a significant negative squared term, but the Lind–Mehlum test for an interior extremum does not pass, the slope at the lower bound of the range being statistically indistinguishable from zero; we therefore report a plateau followed by decline rather than an inverted U. The mechanism is visible in the components: from the first to the second octile mean EGDI rises by 49% while mean penetration rises by 91%, so capacity and reach expand at comparable rates and their product is roughly constant. Decline sets in only as penetration approaches saturation and capacity growth can no longer offset it. The poorest economies do not exhibit the largest facades, because they have not yet built the capacity to strand.

**Stranded service capacity.** Under the narrower truth component of Equation 5, the two facade measures order economies similarly ( $\rho = 0.92$ ) with a mean absolute difference of 0.030, but they diverge systematically, and the divergence is the typology of Section 3.2 measured in  $\Phi$  units. Economies whose online service provision runs ahead of their infrastructure have  $\Phi^{\text{svc}} > \Phi$  — Rwanda by 0.171, Pakistan by 0.158, Uganda by 0.146 — because the EGDI averages their strong service component against weaker infrastructure and human capital. The inverse holds where infrastructure runs ahead: Comoros, Eritrea, and Guinea-Bissau all have substantially less service capacity to strand than their EGDI implies. The two measures are reported in parallel throughout and neither is a correction of the other.

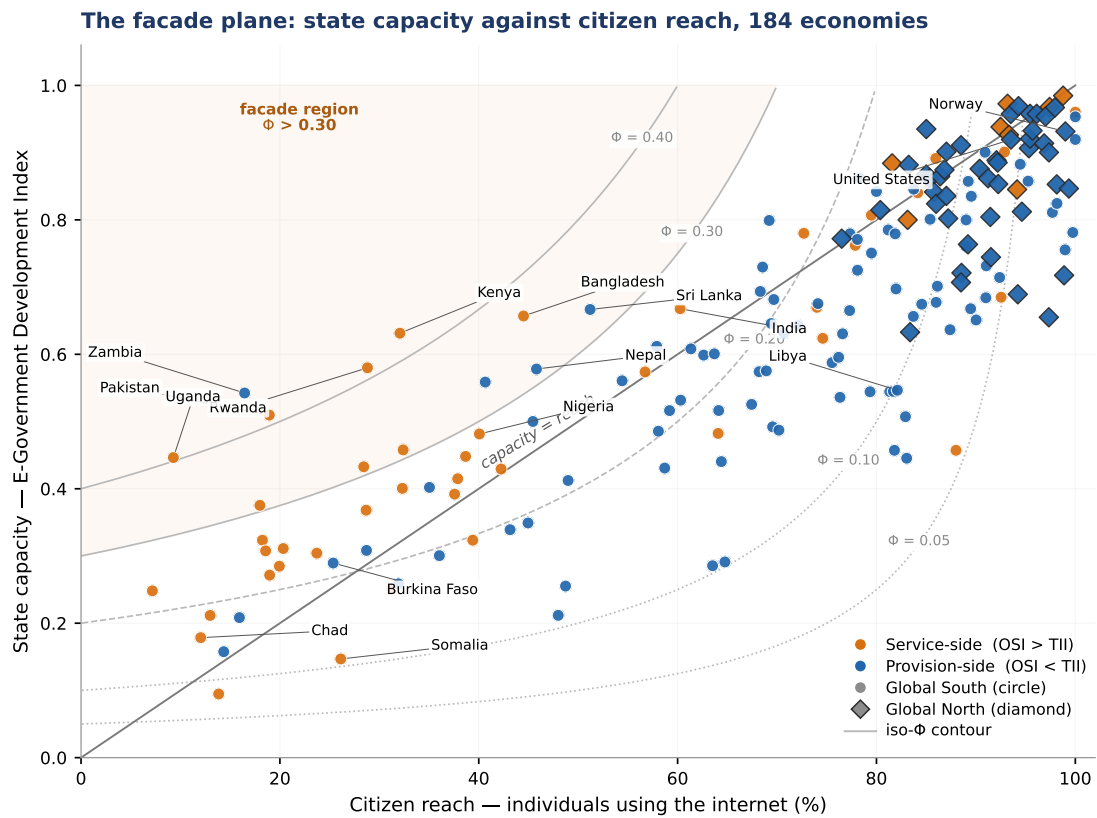


Figure 2: The facade plane. Each economy is placed by citizen reach (horizontal) against state capacity (vertical); grey curves are contours of constant stranded capacity  $\Phi$ , and the diagonal marks parity between capacity and reach. Marker hue distinguishes the two configurations of Section 3.2; diamonds denote economies of the Global North, circles the Global South. The shaded facade region contains twelve economies, none of them Northern.

**Stranded capacity by sub-region — highest in Southern Asia, most widespread in Africa**

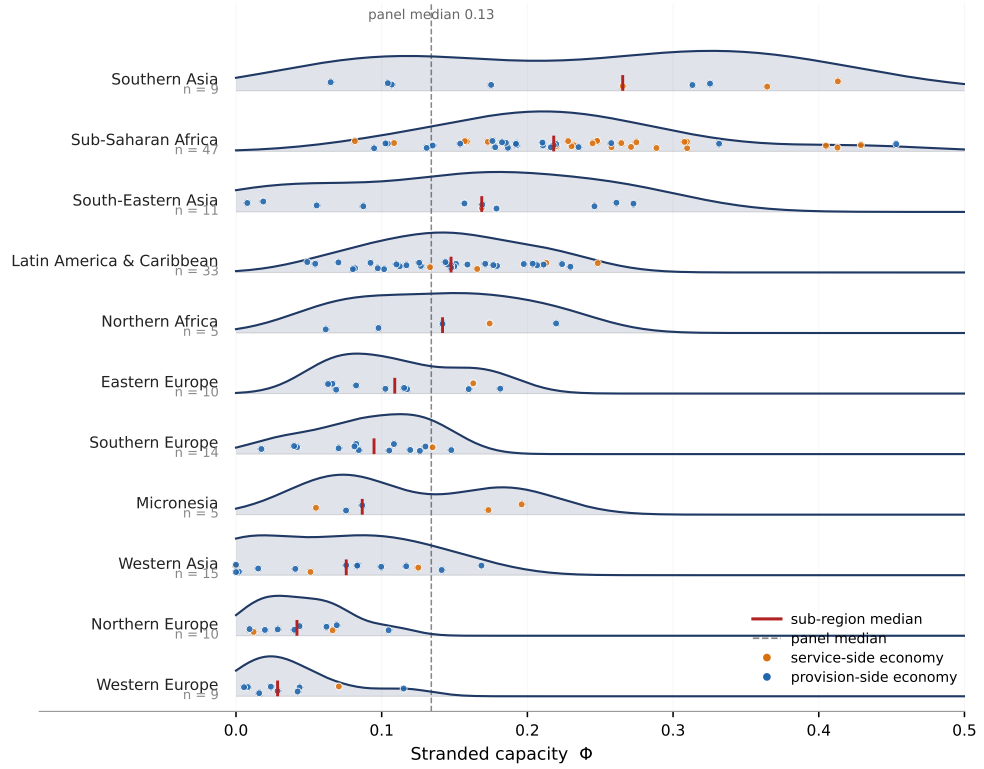


Figure 3: Stranded capacity by M49 sub-region, for sub-regions with at least five economies, ordered by median. Curves are kernel density estimates and points the underlying economies; red rules mark sub-regional medians and the dashed rule the population median. Southern Asia has the highest central tendency on nine economies; sub-Saharan Africa is marginally lower on forty-seven and supplies most of the extreme cases. Densities for the five-economy sub-regions are heavily smoothed and should be read against the points rather than on their own.

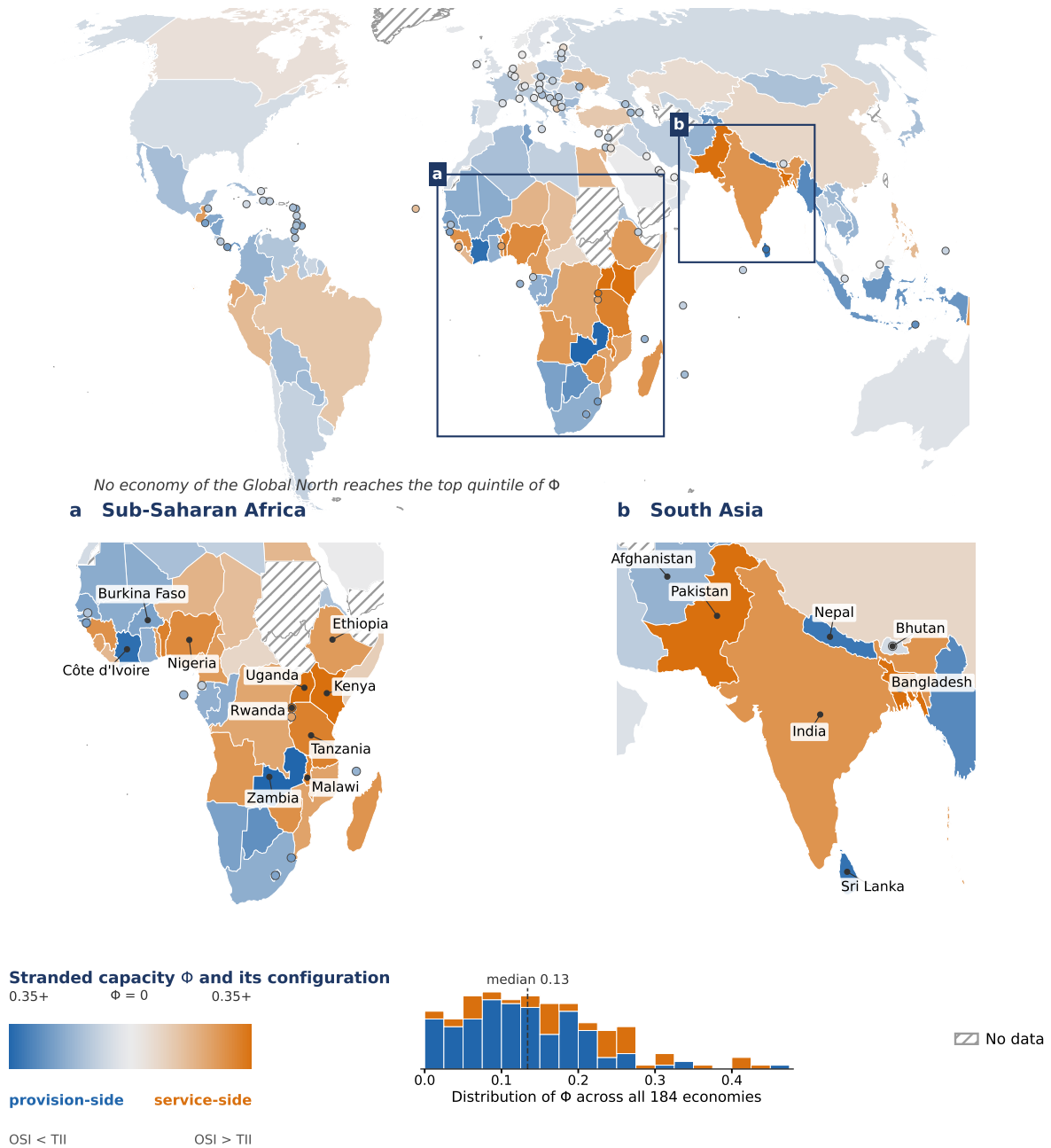


Figure 4: The government facade, 2024. Colour encodes stranded capacity  $\Phi$  jointly with its configuration: hue distinguishes service-side (orange) from provision-side (blue) misalignment, and saturation increases with  $\Phi$ , so economies with little stranded capacity appear near-neutral irrespective of configuration. Hatching marks economies for which an input is unavailable. The two panels magnify the regions in which stranded capacity is concentrated. The histogram gives the distribution of  $\Phi$  across all 184 economies, stacked by configuration. An equal-area projection is used, and economies too small to render as polygons are shown as markers.

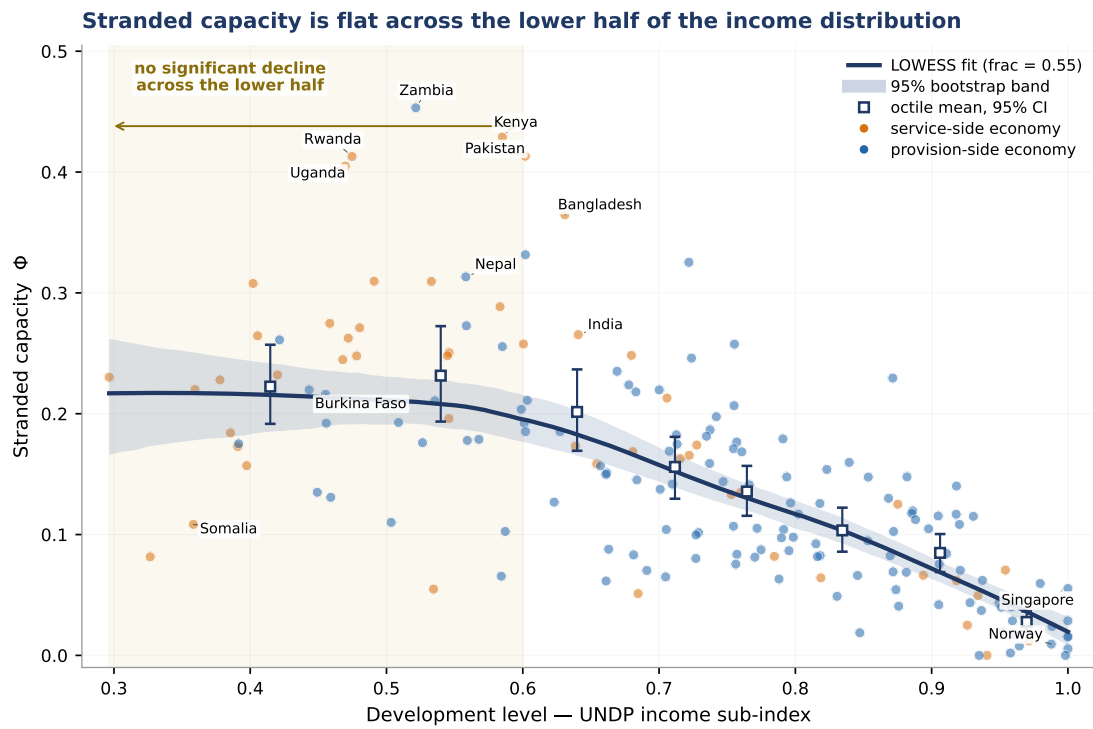


Figure 5: Stranded capacity against development level. The line is a LOWESS fit with a 95% bootstrap band; squares are octile means with bootstrap intervals. The horizontal extent of the shaded region marks the range over which no significant decline is detected. The income sub-index is a log-transformed measure on the UNDP goalpost scale, so the horizontal axis is already logarithmic in gross national income per capita.

### 3.2 The facade typology

The bipolar indeterminacy component of Equation 8 sorts economies into two configurations. Across the Tier I population, 51 economies have  $I_{\text{egov}}^+ > 0$ , online service provision running ahead of the infrastructure carrying it, and 133 have  $I_{\text{egov}}^- < 0$ , the inverse. Provision-side misalignment is therefore the global norm, and by a margin of roughly five to two — or close to four to one among the economies whose divergence is large enough to classify, in the sense set out immediately below.

**The band within which no configuration is asserted.** The counts above classify on the sign of  $D_{\text{egov}}$  alone, and for a substantial minority of economies that sign is smaller than anything the sub-indices resolve. Thirty-four of the 184 economies (18%) fall inside the aligned band of Equation 10,  $|D_{\text{egov}}| \leq 0.05$ ; fourteen fall inside  $|D_{\text{egov}}| \leq 0.02$ ; and Germany’s published divergence is 0.000 to three decimal places, so that its assignment to either pole is nominal. Applying the band, the population divides into 32 service-side, 118 provision-side, and 34 aligned economies. The band is populated by two very different kinds of case — economies whose service and infrastructure profiles genuinely coincide, such as Germany, Estonia, and Kazakhstan, and economies whose profiles are both weak, such as Angola, Cameroon, and the Democratic Republic of the Congo — and the framework distinguishes them by  $\Phi$ , not by the pole. Every claim below is stated on the sign classification, checked against the band, and reported under both where the two could differ.

**The composition inverts among facade cases.** That global ratio reverses where stranded capacity is largest. The service-side share rises monotonically with  $\Phi$ : 28% across the population, 39% above the median, **65%** in the top quintile, and **72%** in the top decile. Economies with the largest facades are predominantly those that have built services ahead of the network required to deliver them — the opposite of the pattern that prevails everywhere else.

The inversion is not an artefact of classifying economies whose divergence is too small to read. Under the aligned band it is if anything sharper, because setting aside the unclassifiable cases removes noise from both ends of the distribution: across the population provision-side economies outnumber service-side ones by 118 to 32, while within the top decile of  $\Phi$  the ratio runs the other way, ten to four, with the remaining four economies inside the band. The direction of the reversal, and its magnitude, therefore do not depend on the treatment of the near-zero cases.

**Configuration tracks development stage.** Service-side economies have a mean income sub-index of 0.608 against 0.767 for provision-side economies ( $p = 2.2 \times 10^{-7}$ , one-sided Mann–Whitney), and a mean  $\Phi$  of 0.197 against 0.125. The service-side share falls from 65% in the poorest income quintile to 31%, 19%, and 8% in the next three, recovering slightly to 16% in the richest. Service-side misalignment is thus concentrated among low-income economies, and it is what facade cases predominantly are.

Figure 6 shows both results in one panel: signed divergence on the horizontal axis, stranded capacity on the vertical, income quintile by marker fill. The upper-right quadrant — large  $\Phi$ , service-side — is occupied almost entirely by economies in the lowest two income quintiles. The

extremes on each side are instructive. Rwanda ( $D_{\text{egov}} = +0.448$ , OSI 0.821 against TII 0.372) and Uganda (+0.377) exhibit the sharpest service-side misalignment in the population and both sit in the top decile of  $\Phi$ . Libya ( $-0.883$ , OSI 0.081 against TII 0.964) is the most extreme provision-side case: near-complete infrastructure carrying almost no online services.

**Why the distinction is not cosmetic.** The two configurations imply opposite interventions. A service-side facade is closed by extending the network and the connectivity that reaches citizens; a provision-side facade is closed by developing and localising the services that run on a network already in place. The aggregate EGDI registers neither, because it averages the two sub-indices whose divergence defines the distinction, and the facade magnitudes of Section 3.1 do not separate them either, since they are conjunctions of truth with falsity. Only the sign of the indeterminacy component carries this information, which is the substantive reason it is specified as bipolar rather than as a magnitude.

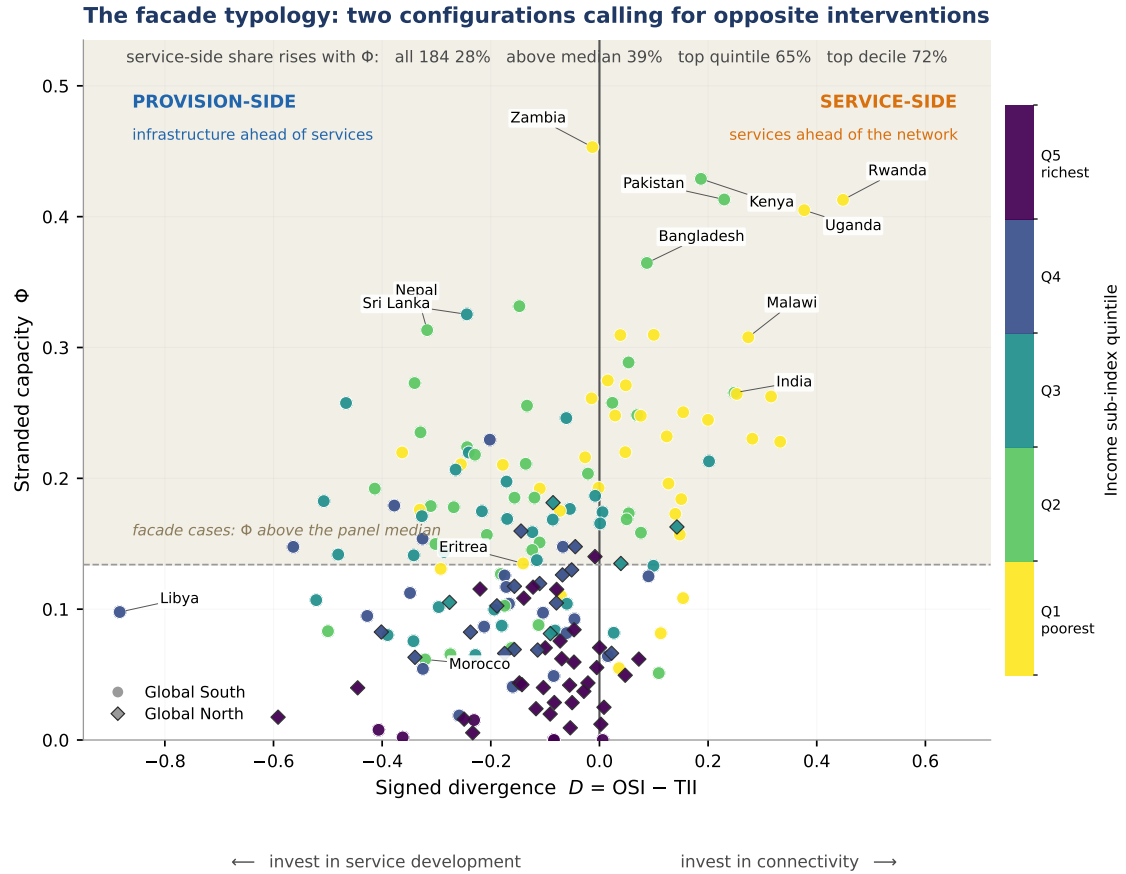


Figure 6: The facade typology. Signed divergence  $D_{\text{egov}} = \text{OSI} - \text{TII}$  against stranded capacity  $\Phi$ , with marker fill giving the income sub-index quintile; circles denote economies of the Global South and diamonds the Global North. Economies above the dashed rule have above-median stranded capacity; the sign on the horizontal axis identifies which configuration each exhibits and therefore which intervention it calls for.

### 3.3 The digital layer

The four digital indicators vary over very different ranges, and the variation is not structured along a single dimension. We describe each in turn on the Tier III population of 158 economies, on which the composite is estimated.

*Individual internet use*  $U$  spans nearly the full unit interval, from **7.2%** in Burundi to saturation in Bahrain, with a median of 81.8%. The distribution is strongly left-skewed: most economies now sit above 70%, and the informative variation is concentrated in the lower tail. This is why access alone is a poor discriminator among developed economies and a decisive one below the median.

*Fixed-broadband subscriptions*  $B$  show a different and less permeable structure, from 0.01 per 100 inhabitants in Timor-Leste to **51.74** in Andorra, with a median of 15.17. Where access has converged, depth of connectivity has not, which is consistent with the argument that infrastructure lag is structurally more persistent than access lag (Van Dijk, 2005). Andorra sets the normalisation base of Equation 2.

*The EGDI* is markedly more compressed than the other three, running from 0.09 in the Central African Republic to **0.98** in Denmark with a median of 0.72. The compression follows from its own composite construction: an economy weak in infrastructure or human capital can still score moderately where online service provision is strong. Unpacking that compression is the work of Sections 3.1 and 3.2.

*High-technology exports*  $H$  have the widest relative range and the sharpest single outlier, from 0.01% in Bhutan to **63.98%** in the Philippines against a median of 6.00%. The Philippine value sets the normalisation base of Equation 3 and reflects assembly and re-export activity rather than domestic innovation (Section 2.2). Because the indicator is a share of manufactured exports, it also runs high where that denominator is narrow, and both distortions are treated as limitations in Section 5.5. Full indicator values for all 184 economies are given in the supplementary materials.

### 3.4 The composite digital sub-index

Table 2 reports the digital sub-index  $\mathbf{I}_{\text{digital}}$  (Equation 15) with its four normalised inputs, for the leading and trailing economies and for four cases that illustrate the aggregator’s behaviour.

Iceland leads at **0.822**, followed by Malta (0.813), Singapore (0.796), and Ireland (0.793). At the floor, Burundi (**0.010**), the Democratic Republic of the Congo (0.011), and the Central African Republic (0.011) sit nearly two orders of magnitude below. That range is wider in relative terms than any of the three standard HDI dimensions, and it is the proximate reason the augmentation displaces economies as far as it does (Section 3.5).

The geometric aggregator refuses to reward a single strong input where others are weak, and three rows make the point concretely. The Philippines attains the population maximum on the normalised high-technology input (**1.000**, as the reference economy for that normalisation) yet ranks 48th at 0.492, because weak broadband (0.126) and a moderate adjusted e-government score (0.593) hold the product down. Kuwait combines near-universal access (0.997) with almost no fixed broadband (0.020) and a negligible high-technology share (0.028), and ranks 113th at 0.144. The Bahamas reports no high-technology exports at all in the reference window; its input

normalises to zero and its composite falls to 0.088 despite moderate performance on the other three. Conversely China reaches 0.712 and tenth position on consistent performance across all four inputs rather than exceptional performance on any one.

This behaviour is a property of the framework rather than an artefact. Equation 15 requires all four inputs to be present and strictly positive, and neither a missing nor a zero input is imputed away, because the substantive claim is that digital capability is genuinely multidimensional and an economy lacking one dimension is not adequately described by a composite that averages over its absence.

Table 2: Digital sub-index  $\mathbf{I}_{\text{digital}}$  (Equation 15) and its four normalised inputs, Tier III ( $N = 158$ ). Column headings abbreviate  $\mathbf{I}_{\text{penetration}}$ ,  $\mathbf{I}_{\text{broadband}}$ , and  $\mathbf{I}_{\text{hitech}}$ . The five leading and four trailing economies are shown, with four intermediate cases discussed in the text. Population extrema are set in bold. Full values are given in the supplementary materials.

| Economy                          | $\mathbf{I}_{\text{pen}}$ | $\mathbf{I}_{\text{bb}}$ | $T_{\text{egov}}^{\text{adj}}$ | $\mathbf{I}_{\text{ht}}$ | $\mathbf{I}_{\text{digital}}$ | Rank |
|----------------------------------|---------------------------|--------------------------|--------------------------------|--------------------------|-------------------------------|------|
| Iceland                          | 0.979                     | 0.723                    | 0.947                          | 0.679                    | <b>0.822</b>                  | 1    |
| Malta                            | 0.921                     | 0.856                    | 0.818                          | 0.676                    | 0.813                         | 2    |
| Singapore                        | 0.943                     | 0.530                    | 0.914                          | 0.877                    | 0.796                         | 3    |
| Ireland                          | 0.969                     | 0.616                    | 0.885                          | 0.747                    | 0.793                         | 4    |
| Republic of Korea                | 0.974                     | 0.900                    | 0.943                          | 0.473                    | 0.791                         | 5    |
| China                            | 0.906                     | 0.865                    | 0.790                          | 0.415                    | 0.712                         | 10   |
| Philippines                      | 0.779                     | 0.126                    | 0.593                          | <b>1.000</b>             | 0.492                         | 48   |
| Kuwait                           | 0.997                     | 0.020                    | 0.779                          | 0.028                    | 0.144                         | 113  |
| Bahamas                          | 0.924                     | 0.460                    | 0.660                          | 0.000                    | 0.088                         | 129  |
| Timor-Leste                      | 0.350                     | 0.000                    | 0.141                          | 0.024                    | 0.019                         | 155  |
| Central African Republic         | 0.138                     | 0.000                    | 0.013                          | 0.029                    | 0.011                         | 156  |
| Democratic Republic of the Congo | 0.190                     | 0.000                    | 0.051                          | 0.003                    | 0.011                         | 157  |
| Burundi                          | 0.072                     | 0.000                    | 0.018                          | 0.020                    | <b>0.010</b>                  | 158  |

### 3.5 The HDI-N composite and rank reordering

HDI-N (Equation 16) is defined for the 158 Tier III economies. Because the standard HDI is published as a global rank over roughly 190 economies, the two are not directly comparable as published: the economies absent from Tier III occupy rank positions and displace every economy below them. Both rankings reported here are therefore computed within the same 158 economies, and the standard-HDI rank quoted below will not match the rank published in the Human Development Report.

**Correlation and displacement.** The correlation test specified in Section 2.8 returns Spearman  $\rho = \mathbf{0.947}$  against the standard HDI, within the intermediate band fixed in advance: high enough that the augmentation does not override the underlying human development structure, far enough from unity that it is not a relabelling.

The correlation alone understates what the augmentation does, because rank correlation is high whenever the underlying range is wide, and this range spans the full development distribution. Displacement is the more informative statistic. Of the 158 economies, **122 (77%)** move at least three rank positions, **101 (64%)** move at least five, and **65 (41%)** move at least ten.

The largest single displacement is **62 positions**. A high rank correlation and a substantial re-ordering are not in tension; Figure 7 shows both at once, as a tight diagonal with a wide scatter around it.

**What the augmentation rewards and penalises.** The displacement is structured, and the structure is interpretable. Table 3 reports the leading economies together with the largest movers in each direction.

Economies that gain are those whose digital capability runs ahead of their human-development position: Viet Nam (+41), the Philippines (+37), China (+32), and Mongolia (+28). Each combines a middling standard-HDI rank with a digital sub-index well above what that rank would predict — Viet Nam at 0.614 and China at 0.712, against a Tier III median of 0.306.

Economies that fall share a common profile that the twenty-economy panel did not contain: small, high-income, and resource- or tourism-dependent. The Bahamas loses **62** positions, Kuwait 54, Seychelles 45, Bahrain 42, Trinidad and Tobago 40, and Panama 35. Their digital sub-indices are among the lowest in the population — the Bahamas at 0.088, Kuwait at 0.144 — against standard-HDI ranks in the top third. High income has not translated into digital capability in these economies, and the geometric aggregator penalises that imbalance heavily. This is the same mechanism the UNDP invoked in 2010 when it moved from arithmetic to geometric aggregation, here acting on a dimension the standard index does not contain.

At the head of the ranking, Iceland leads at 0.923, followed by Switzerland (0.913) and Singapore (0.908). The Republic of Korea rises eleven positions to sixth and Malta fifteen to eighth; Norway falls three. The digital sub-index itself spans nearly two orders of magnitude, from 0.822 (Iceland) to 0.010 (Burundi), which is a wider relative range than any of the three standard dimensions and is why the fourth dimension moves the composite as much as it does.

### 3.6 External validation against demand-side survey data

The three preceding subsections describe a quantity built from published indices. This one asks whether it corresponds to anything citizens actually experience. The exercise is the eighth item of the protocol (Section 2.8); it is reported here rather than with the robustness results because it is a finding about the world and not a sensitivity result about the framework.

**The state-mediated indicator.** Regressing the share of adults receiving government payments into an account on the EGDI and the UNDP income sub-index, and then adding  $\Phi$ , the coefficient on stranded capacity is **-0.462** (robust standard error 0.109,  $p = 2.3 \times 10^{-5}$ ,  $N = 145$ ), and  $R^2$  rises from 0.592 to 0.624. Where more of a state’s digital capacity is stranded, fewer citizens receive government payments digitally than that state’s capacity and income level would predict. The same holds for use of a mobile phone or the internet to reach an account ( $-0.443$ ,  $p = 0.006$ ,  $N = 88$ ). The result survives substituting  $\Phi^{\text{svc}}$  for  $\Phi$  ( $-0.278$ ,  $p < 0.001$ ) and restricting the sample to the Global South, where the range of  $\Phi$  is widest ( $-0.360$ ,  $p < 0.001$ ,  $N = 97$ ).

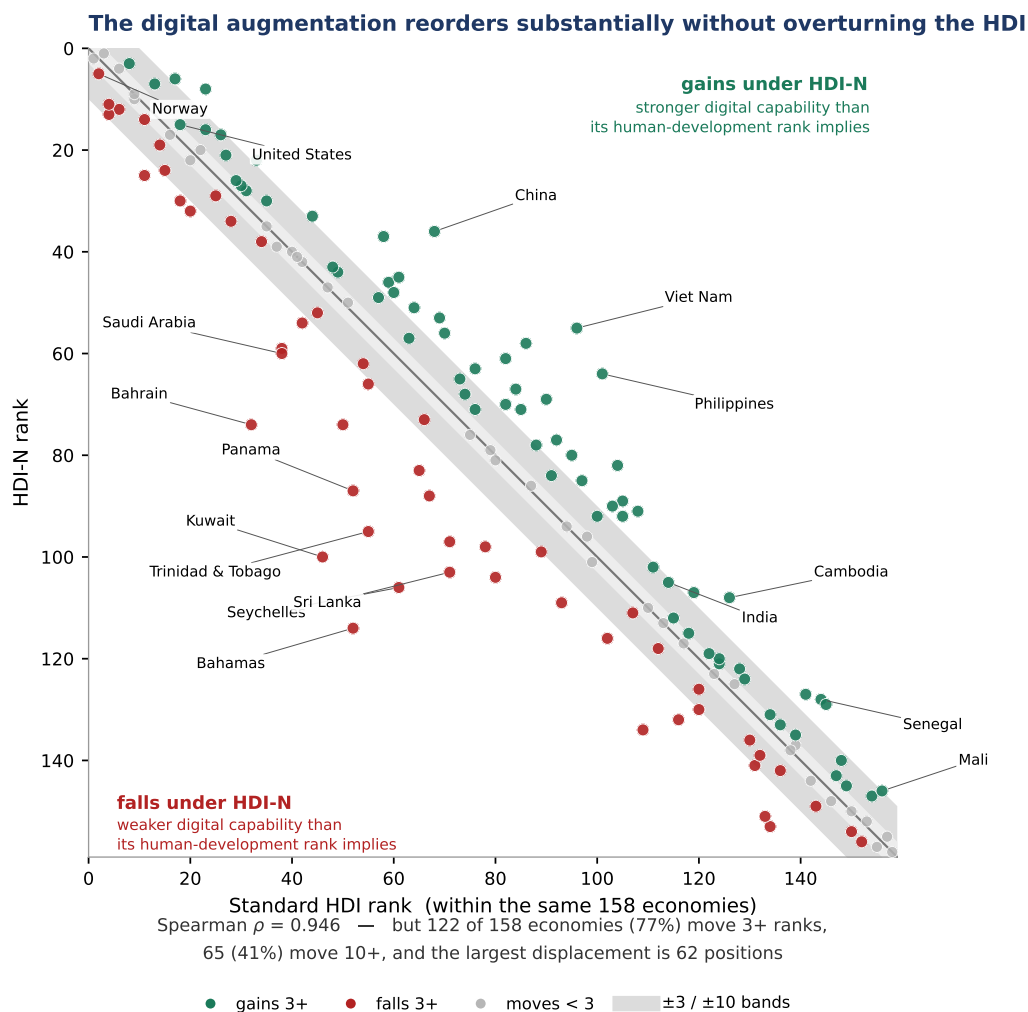


Figure 7: HDI-N rank against standard HDI rank, Tier III ( $N = 158$ ). Both ranks are computed within the same 158 economies. Shaded bands mark displacements of  $\pm 3$  and  $\pm 10$  positions. Economies above the diagonal gain under the augmentation, those below it fall. The Spearman correlation of 0.947 and the displacement of 77% of economies by three or more positions are properties of the same scatter.

Table 3: HDI-N and rank comparison with the standard HDI, Tier III ( $N = 158$ ). The eight leading economies are shown, followed by the four largest gainers and the four largest losers. Both ranks are computed within the 158 economies for which HDI-N is defined, so the standard-HDI rank does not match the published global rank.  $\Delta$  is positive where the HDI-N rank is higher. The full ranking is given in the supplementary materials.

| Economy                  | HDI-N | Rank | Std. HDI | Rank | $\Delta$ |
|--------------------------|-------|------|----------|------|----------|
| <i>Leading economies</i> |       |      |          |      |          |
| Iceland                  | 0.923 | 1    | 0.959    | 3    | +2       |
| Switzerland              | 0.913 | 2    | 0.967    | 1    | -1       |
| Singapore                | 0.908 | 3    | 0.949    | 8    | +5       |
| Ireland                  | 0.908 | 4    | 0.950    | 6    | +2       |
| Norway                   | 0.908 | 5    | 0.966    | 2    | -3       |
| Republic of Korea        | 0.892 | 6    | 0.929    | 17   | +11      |
| United Kingdom           | 0.888 | 7    | 0.940    | 13   | +6       |
| Malta                    | 0.888 | 8    | 0.915    | 23   | +15      |
| <i>Largest gains</i>     |       |      |          |      |          |
| Viet Nam                 | 0.696 | 55   | 0.726    | 96   | +41      |
| Philippines              | 0.647 | 64   | 0.710    | 101  | +37      |
| China                    | 0.769 | 36   | 0.788    | 68   | +32      |
| Mongolia                 | 0.669 | 58   | 0.741    | 86   | +28      |
| <i>Largest falls</i>     |       |      |          |      |          |
| Bahamas                  | 0.470 | 114  | 0.820    | 52   | -62      |
| Kuwait                   | 0.543 | 100  | 0.847    | 46   | -54      |
| Seychelles               | 0.494 | 106  | 0.802    | 61   | -45      |
| Bahrain                  | 0.617 | 74   | 0.888    | 32   | -42      |

**The discriminant criterion is met.** One indicator shows nothing, and its silence carries as much weight as the others’ significance. Broad digital payment activity — any digital payment made or received, including peer-to-peer transfers and private mobile money — is unrelated to  $\Phi$  ( $-0.092$ ,  $p = 0.71$ ). Inspection of the residuals shows why: Kenya, Uganda, Tanzania, and Zambia combine large stranded capacity with exceptionally high digital payment uptake, the signature of mobile-money ecosystems that operate independently of state provision.  $\Phi$  therefore predicts *state-mediated* digital uptake specifically, and not digital activity in general. This is the discriminant result the first criterion of Section 2.8 required: a proxy for connectivity or for development would have moved the broad indicator too.

**The incremental criterion.** The second and more demanding criterion asks whether the conjunction outperforms the falsity component whose variance it inherits. Entering the EGDI, the income sub-index,  $F_{\text{egov}}$ , and  $\Phi$  in a common specification gives the following.

[PENDING: insert the horse-race estimates here — coefficient, robust standard error,  $p$ -value and  $N$  for  $\Phi$  and for  $F_{\text{egov}}$  in the common specification, for the state-mediated indicator and for the broad indicator, together with the  $R^2$  against the specification that omits  $\Phi$ . Then replace this paragraph with the reading the criterion in Section 2.8 licenses, whichever way the estimates fall.]

The component-level evidence in Section 3.1 bears on the same question from the other direction: the two quantities select substantially different sets of economies at the extremes, which is where the measure is used. The regression asks whether that difference is also visible

in outcomes.

## 4 Robustness

The seven sensitivity checks specified in Section 2.8 are reported here in order; the eighth item of that protocol, the external validation, is a finding rather than a sensitivity result and was reported in Section 3.6. Taken together the seven establish that the paper’s substantive claims are stable across measurement-noise assumptions, aggregation-operator choice, and the depth of the neutrosophic aggregation; that the digital augmentation reorders countries without overriding the underlying human development structure; and that the largest reorderings are attributable to a specific and identifiable channel.

### 4.1 Bootstrap perturbation across noise levels

Every normalised sub-indicator was perturbed additively (Equation 22) over 1,000 replicates at each of three noise levels, and the HDI-N composite recomputed. Median ranks are identical to the unperturbed ranking at all three levels: the median absolute difference between the perturbed median rank and the base rank is zero throughout, so no economy’s central position is an artefact of measurement noise.

What changes with the noise level is interval width, and it widens as expected. At  $\tau = 0.01$  the mean 95% interval spans 2.8 rank positions and 84% of economies have an interval of four positions or fewer; at the reference level  $\tau = 0.03$  the mean width is 7.6 positions; at  $\tau = 0.05$  it is 12.1. Even at the highest level the maximum width is 19 positions out of 158, so the ordinal structure is preserved across the plausible range of measurement-noise assumptions.

Two features of the intervals warrant comment. The top of the ranking is close but stable: Iceland has median rank 1 with an interval of [1, 6], and Switzerland, Ireland, Singapore, and Norway all have medians between 3 and 4 with intervals contained in [1, 9]. Their HDI-N scores are separated by less than 0.02, so perturbation exchanges them freely without any of them leaving the leading group. The bottom of the ranking is by contrast fully determined: Madagascar, Niger, and the Democratic Republic of the Congo have zero-width intervals at  $\tau = 0.03$ , and Burundi and the Central African Republic differ only in which occupies the final position. Where the composite is near its floor, no plausible perturbation displaces an economy.

Sub-indicators are perturbed independently. This is a disclosed simplification: internet penetration enters the composite twice, directly and inside  $T_{\text{egov}}^{\text{adj}}$  (Equation 11), so a fully specified noise model would perturb the two entries in a correlated fashion. Independent perturbation therefore understates the effective variance attaching to penetration, which is one reason the protocol reports a ladder of noise levels rather than a single value.

### 4.2 Sensitivity to aggregation operator

The composite was recomputed under the arithmetic mean and under the CES operator at  $\theta = -1$ , the harmonic mean (Equation 23). Spearman rank correlations across the 158 economies are 0.988 (geometric–arithmetic), **0.996** (geometric–harmonic), and 0.972 (arithmetic–harmonic). All exceed 0.97: the country ordering is not an artefact of the geometric mean and would be

substantially reproduced under either alternative. Cross-operator comparison of this kind is standard practice — Chakrabarty & Bhattacharjee (2012), for a digital composite over Asian economies, cross four weighting schemes with linear, geometric, and displaced-ideal aggregation and compare by rank correlation.

The individual movements are informative, and they concentrate where the choice of aggregator should matter most. Fifty-six economies move five or more positions between the geometric and arithmetic orderings, with a maximum displacement of 33; the economies that move furthest are exactly those with the most unbalanced digital profiles. Kuwait rises 33 positions under arithmetic aggregation and falls a further 10 under the harmonic; the Bahamas, Bahrain, Timor-Leste, and Bhutan behave the same way. Each combines one or two strong inputs with at least one near-zero input, and the ordering of the three aggregators — arithmetic most permissive, geometric intermediate, harmonic most punitive — reproduces the imbalance penalty exactly. This is a direct empirical illustration of the property that motivated the UNDP’s 2010 shift (UNDP, 2010; Klugman et al., 2011), here in the digital dimension specifically.

The comparison also bears on the normalisation choice of Section 2.3, and more strongly than the invariance argument given there. The geometric composite is exactly invariant to the divisors of Equations 2 and 3; the arithmetic composite is not. Rank stability under changes of normalisation is therefore not a generic property of composite construction but a consequence of multiplicative aggregation specifically — an argument for the geometric mean independent of the imbalance-penalising rationale inherited from UNDP practice.

### 4.3 Correlation between HDI-N and the standard HDI

Across the 158 economies for which HDI-N is defined, the Spearman rank correlation with the standard HDI is  $\rho = \mathbf{0.947}$ . Against the thresholds fixed in Section 2.8, this is the intended result: a value near unity would indicate that the augmentation does no analytical work, and one substantially below 0.8 that it overrides the underlying HDI structure.

The interpretation of that figure requires one caution, which the twenty-economy version of this analysis did not face. Rank correlation rises mechanically with the range of the underlying variable, and the present population spans the full development distribution rather than a selected band. A correlation of 0.947 is therefore compatible with substantial reordering, and Section 3.5 establishes that substantial reordering is what occurs: 77% of economies move at least three positions and 41% move at least ten. We accordingly report displacement alongside correlation throughout, since neither statistic alone characterises what the augmentation does.

### 4.4 Full-triple neutrosophic aggregation

Carrying the complete triple  $(T_{\text{egov}}, I_{\text{egov}}, F_{\text{egov}})$  through the pipeline under the multiplicative extension (Equation 24) and under the score function (Equation 25) reproduces the main-text ordering almost exactly. Spearman correlations against the two-component ordering are **0.9994** and **0.9993** respectively, and the two full-triple operators correlate with each other at 0.9990 — despite embodying opposite assumptions about how the components combine, one conjunctive and one additive.

Displacement is confined and, where it occurs, is signed in the direction the operators predict. Sixteen economies move three or more positions between the two-component and multiplicative orderings and the largest displacement is seven positions out of 158. Every economy with  $I_{\text{egov}} > 0.35$  — twelve of the 158 — either falls or holds under the multiplicative operator, and none rises: Brunei Darussalam and Andorra fall seven positions each, Iran and Montenegro five. Elsewhere movement is one or two positions and unsigned, which is the mechanical consequence of other economies moving. The second criterion of Section 2.8 is therefore met: the economies the operators treat differently are those whose indeterminacy is largest. Figure 8 shows both results.

The mechanism is the double geometric damping of Equations 15 and 16. Because  $T_{\text{egov}}^{\text{adj}}$  enters HDI-N through two nested fourth roots, a proportional discount  $d$  applied to it rescales the composite by  $(1 - d)^{1/16}$ ; a discount of 40% therefore moves the composite by about 3%. The two-component operator of Equation 11 is accordingly retained in the main text as the parsimonious aggregation whose ordering the complete triple reproduces, satisfying the criterion set in Section 2.8. Full-triple scores are reported in the supplementary materials.

#### Ordinal inertness of the neutrosophic triple (Tier III, $N = 158$ )

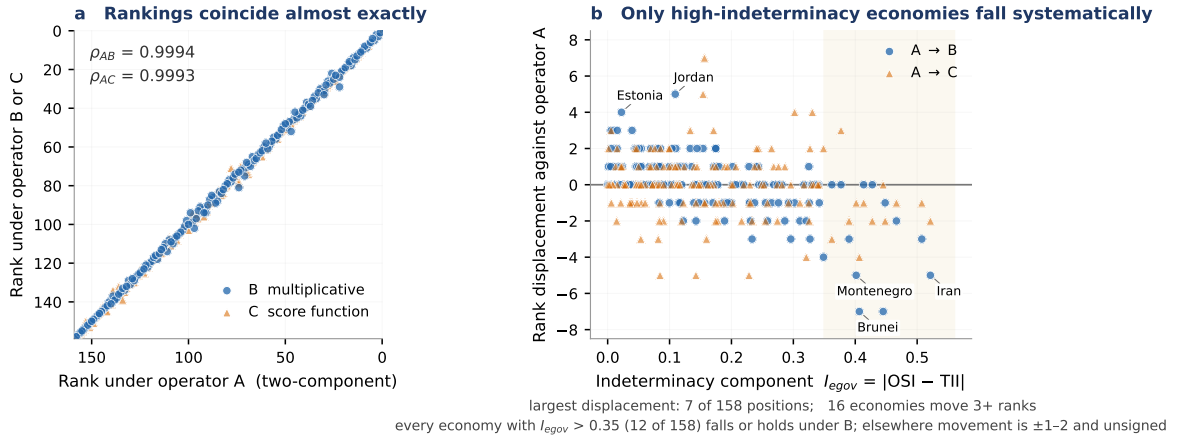


Figure 8: Ordinal inertness of the neutrosophic triple, Tier III ( $N = 158$ ). (a) Rank under the multiplicative extension (operator B) and under the score function (operator C), each plotted against rank under the two-component operator (A); the shaded band marks  $\pm 3$  positions. (b) Rank displacement against operator A, plotted against the magnitude of the indeterminacy component. The shaded region marks  $I_{\text{egov}} > 0.35$ , within which every economy falls or holds under the three-component operators and none rises.

#### 4.5 Sensitivity to the composition of the digital sub-index

The digital sub-index (Equation 15) raises two questions. One input, high-technology exports, measures the economic conversion of digital capability into tradeable output rather than citizen or state digital capacity directly; and internet penetration enters twice, once directly and once inside  $T_{\text{egov}}^{\text{adj}}$ . Both are addressed by a  $2 \times 2$  grid crossing the composition of the sub-index (four inputs; three inputs, omitting high-technology exports) with the treatment of e-government capacity (the adjusted score; the raw EGDI, which removes the second entry of penetration).

The two axes behave very differently, and the contrast is itself the substantive result.

*The treatment of e-government capacity is ordinally inert.* Replacing  $T_{\text{egov}}^{\text{adj}}$  with the raw EGDI leaves the four-input ordering essentially unchanged: Spearman  $\rho = \mathbf{0.9994}$ , with 16 of 158 economies moving three or more positions and a maximum displacement of five. The three-input variants agree at  $\rho = 0.9991$ . The double entry of penetration is therefore consequential for the level of the sub-index but not for the ordering, and the main-text specification is retained on interpretive grounds at negligible ordinal cost.

*The composition of the sub-index is decisive.* Omitting high-technology exports moves the ordering substantially:  $\rho = 0.963$  against the four-input specification, with 105 economies moving three or more positions and a maximum displacement of **60**. The rank movement between the four- and three-input variants therefore isolates the contribution of the conversion channel, and the economies it moves furthest are precisely those whose high-technology export share is near zero or extreme. The Bahamas rises **60** positions when the channel is removed, Seychelles 50, Trinidad and Tobago 38, Saudi Arabia 37, and Bahrain 35 — each reporting a high-technology share below 0.6%. In the opposite direction, the Philippines falls 37 positions and Malaysia 26, both on shares above 59%.

That contrast identifies what the conversion channel measures and what it does not. It is a merchandise-export construction, so it registers assembly and re-export activity as conversion and registers its absence in economies whose exports are services, tourism, or primary commodities. The three-input variant is a diagnostic for locating where rank movement originates, not an alternative measure: its correlation with the standard HDI is 0.981 against 0.947 for the four-input specification, which is to say that removing the conversion channel makes the composite substantially more like the index it augments. The four-input specification is retained, following Daghigh Yazd et al. (2025), on the position that economic conversion of digital capability is constitutive of the digital dimension rather than incidental to it. Its limitations as a goods-only measure are treated in Section 5.5.

#### 4.6 Sensitivity of the facade characterisation to Pakistan’s penetration vintage

Pakistan’s internet use is carried forward from the 2020 ITU release (18.94%); the DataHub reports no values for 2021–2023, a gap re-verified against the source on 27 July 2026, while the 2024 release reports 57.25%. A discontinuity of this size reflects revised measurement rather than organic growth, and the within-window gap prevents the audit protocol of Section 2.2 from resolving it. This check treats the two figures as interval endpoints and asks which claims are sensitive to the choice.

Under the audited carry-forward, Pakistan’s stranded capacity is 0.413, third of 184, on a facade ratio of 5.28. Under the 2024 release value as a nowcast,  $\Phi$  falls to 0.218 and the ratio to 1.75, placing Pakistan 40th of 184 — inside the top quartile but outside the top decile.

The sensitivity is therefore real but bounded, and its consequences differ by claim. Pakistan’s individual position is vintage-dependent and no substantive claim in this paper rests on it. The claims that do rest on the distribution are unaffected: the sub-Saharan concentration of Section 3.1 does not involve Pakistan; the Mann–Whitney test on the North–South partition

returns the same conclusion at either endpoint, Pakistan being a Southern economy in both cases; and the service-side character of Pakistan’s misalignment ( $D_{\text{egov}} = +0.230$ ) is a property of the EGDI sub-indices alone and is invariant to the penetration figure entirely. The audited value is retained in the main analysis for consistency with the reference-year policy of Section 2.2, and Pakistan is reported throughout as an economy whose facade magnitude is bounded below by the nowcast.

#### 4.7 Sensitivity of the regional finding to the cutoff

The regional claims of Section 3.1 are made on the continuous distribution of  $\Phi$ , but a reader may reasonably ask whether they depend on that choice. Applying a top- $k$  rule to  $\Phi$  at three cutoffs gives a sub-Saharan African share of the selected set of 67% at the decile ( $k = 18$ , Fisher exact  $p = 1.3 \times 10^{-4}$ ), 62% at the quintile ( $k = 37$ ,  $p = 9.2 \times 10^{-8}$ ), and 59% at the quartile ( $k = 46$ ,  $p = 1.7 \times 10^{-8}$ ), against a base rate of 26%. The concentration holds at every cutoff tested.

The continuous test is nonetheless the stronger one, and materially so: the Mann–Whitney comparison on  $\Phi$  itself returns  $p = 2.3 \times 10^{-12}$ , four orders of magnitude smaller than the decile-based test, because it uses the full distribution rather than thresholding it away. A joint rule on  $\Phi$  and  $R_{\text{facade}}$  together is more restrictive again and less stable: the two quantities correlate at  $\rho = 0.90$ , so their joint top decile contains only five economies, and membership at that cut is sensitive to displacement of a single rank. We therefore report the continuous quantities throughout and treat this section as establishing that nothing turns on that decision.

## 5 Discussion

We take the results in three parts: what the facade measure is and what evidence there is that it measures what it claims; what the typology adds and why its composition among facade cases is theoretically interesting; and what the digital augmentation does to the human development ranking. We then turn to limitations and future work.

### 5.1 The facade is measurable, and the measure has external support

The decomposition gives quantitative form to a long-standing observation in the e-government implementation literature. Heeks (2002) framed the issue as a *design–reality gap*: a mismatch between what e-government systems are built to do and what local conditions allow them to accomplish. That literature diagnosed the gap without measuring it. Stranded capacity measures it, and does so on 184 economies from two published series.

The measure is not, however, self-validating. A quantity constructed as the product of two indices can be arithmetically transparent and still correspond to nothing observed, and no amount of internal robustness testing can distinguish the two cases. Section 3.6 therefore tested  $\Phi$  against evidence drawn from outside the indices that construct it — the demand-side household survey of the Global Findex Database, which records what citizens do rather than what states are assessed to provide — against criteria fixed in advance in Section 2.8. Three features of that result deserve comment here.

The first is what the significant coefficients mean substantively. Conditional on state capacity and income level, an economy at the top of the  $\Phi$  distribution reaches materially fewer of its citizens with digital government payments than an economy of the same measured capacity and the same income whose provision is within reach. The gap is not a residual left over from a poorly specified model; it is what the facade *is*, and it is visible in behaviour rather than only in the indices.

The second is that the null result on broad digital payment activity is the more informative of the two. A measure that is really a proxy for connectivity or for development would depress digital activity of every kind. This one does not: mobile-money ecosystems in East and Southern Africa run at high intensity in exactly the economies where stranded capacity is largest, and  $\Phi$  has nothing to say about them. Discriminant results of this shape are the hardest for a constructed measure to obtain by accident, and their absence is the most common reason composite indicators fail to persuade.

The third is the limit of the exercise. The validation is cross-sectional and correlational; it establishes correspondence between a constructed quantity and an independently surveyed outcome, not that stranded capacity causes low uptake. Reverse causation is implausible in the direction that would matter — low uptake does not build e-government portals — but omitted determinants of both, such as administrative reach or trust in the state, are not controlled and would need a panel to address.

## 5.2 Two configurations, and what their composition implies

The bipolar indeterminacy component separates two configurations that a single facade magnitude cannot distinguish, and the distinction is directly actionable. A service-side facade — online provision built ahead of the network carrying it — calls for connectivity investment, spectrum allocation, and affordability measures. A provision-side facade — infrastructure running ahead of the services deployed on it — calls for service development, digitisation of administrative processes, and localisation. The aggregate EGDI registers neither, because it averages the two sub-indices whose divergence defines the distinction.

The composition of the two types is the more interesting result. Across the population, provision-side misalignment predominates by roughly five to two. Among facade cases it inverts: 65% of the top quintile of  $\Phi$  and 72% of the top decile are service-side, and the reversal holds in the same direction and to a similar degree when the economies whose divergence falls below the resolution of the sub-indices are set aside (Section 3.2). Economies with the largest facades are predominantly those that have built the *assessed* component ahead of the physical one.

That asymmetry invites a mechanism, and one is available. The two EGDI sub-indices are not symmetrically observable. The Online Services Index is scored from an assessment of national portals conducted against a questionnaire; the Telecommunications Infrastructure Index is computed from ITU telecommunications statistics. One measures what a government presents to an assessor, the other what its network physically carries. A state that responds to international assessment will find the first far easier to move than the second. This is the pattern Pritchett et al. (2013) term *isomorphic mimicry*: the adoption of the observable forms of capable institutions ahead of their function, sustained because the form is what external

evaluation rewards. The EGDI is precisely such an evaluation, published biennially and used in international comparison, and our finding is that the economies furthest from translating capacity into citizen benefit are disproportionately those whose capacity is concentrated in its assessed component.

We state the limits of this reading. The pattern is consistent with isomorphic mimicry but does not establish it: the data are cross-sectional, and an alternative explanation — that online services are simply cheaper to build than networks, so any capacity-constrained state builds them first — would produce the same cross-sectional signature. Distinguishing the two requires the time-series application proposed in Section 5.6, since mimicry predicts that the service–infrastructure gap responds to assessment cycles while the cost explanation does not. What the present data establish is the asymmetry itself, its concentration among low-income economies, and its association with the largest facades.

### 5.3 Facade is a developmental stage, not a poverty attribute

Stranded capacity is flat across the lower half of the income distribution and declines steeply thereafter (Section 3.1). The poorest economies do not exhibit the largest facades. Burundi, with the highest facade ratio in the population, has an EGDI of 0.25; Somalia and Chad sit low on stranded capacity for the same reason. A facade requires capacity to strand, and the poorest economies have not yet built it.

This distinguishes the phenomenon from underdevelopment as such and gives it a temporal reading. Capacity and reach both expand with income, and through the lower half of the distribution they expand at comparable rates, so their product is roughly constant. Decline sets in only as penetration approaches saturation, at which point capacity growth can no longer be offset. The facade is therefore a transitional configuration: something a state passes through as it builds digital capacity faster than its population acquires the means to use it.

We resisted a stronger claim the data do not support. A quadratic specification returns a significant negative squared term, which would conventionally be read as an inverted U with a turning point in the lower income range. The Lind and Mehlum (2010) test for an interior extremum does not pass, the slope at the lower bound of the observed range being statistically indistinguishable from zero, and the curvature is carried by eleven economies at the boundary. We therefore report a plateau followed by decline. The distinction matters for what follows from it: a turning point would imply that facades emerge at a specific developmental stage, whereas a plateau implies only that they do not systematically worsen as income falls.

### 5.4 Digital augmentation and the reordering of human development

HDI-N reorders the human development ranking substantially without overturning it: Spearman  $\rho = 0.947$  against the standard index, with 77% of economies moving three or more positions and 41% moving ten or more. Those two statistics are compatible, and reporting only the first would understate what the augmentation does.

The reordering is structured. Economies that gain — Viet Nam (+41), the Philippines (+37), China (+32), Mongolia (+28) — combine a middling human-development position with digital capability well above what that position predicts. Economies that fall share a profile: small,

high-income, and dependent on resources or tourism. The Bahamas falls 62 positions, Kuwait 54, Seychelles 45, Bahrain 42. Their digital sub-indices are among the lowest in the population despite standard-HDI ranks in the top third.

This is the imbalance penalty that motivated the UNDP’s 2010 shift from arithmetic to geometric aggregation (UNDP, 2010; Klugman et al., 2011), acting on a dimension the standard index does not contain. Whether it should so act is a substantive question rather than a technical one, and we state our position: if digital capability is constitutive of contemporary human capability, then an economy that has converted income into health, education, and longevity but not into digital capability has an imbalance that a development measure ought to register. The alternative position — that digital capability is instrumental rather than constitutive, and should therefore not penalise — is defensible, and it implies additive rather than geometric aggregation. Section 4.2 reports the ranking under that assumption.

One channel deserves separate treatment because it drives much of the movement. High-technology exports enter as a share of manufactured exports, and removing that input moves the ranking substantially (Section 4.5), lifting the Bahamas 60 positions and Seychelles 50. The measure registers assembly and re-export activity as conversion, and registers its absence in economies whose exports are services or commodities. The three-input variant correlates with the standard HDI at 0.981 against 0.947 for the four-input specification: removing the conversion channel makes the composite substantially more like the index it augments, which is a reason to retain it and a reason to interpret it narrowly, as a statement about merchandise export structure rather than about digital conversion in the round.

## 5.5 Limitations

**The components are not fully independently sourced.** The Telecommunications Infrastructure Index is built in part from ITU internet-user data, so  $T_{\text{egov}}$  is not fully independent of the series supplying  $F_{\text{egov}}$  (Section 2.4). The direction of the resulting bias is determinate and conservative: stranded capacity is attenuated precisely where non-reach is largest. The indeterminacy component is computed from the EGDI’s own sub-indices and is therefore a within-index diagnostic rather than an externally sourced component. No cross-country source supplies an externally measured indeterminacy for e-government capacity, and constructing one is left to future work.

**Coverage declines with analytical depth.** The facade analysis covers 184 economies, the composite 158, and the difference is set entirely by high-technology export availability (Section 2.2). The exclusions are not random: they fall on economies with a narrow manufacturing base, and they include Bangladesh, which ranks sixth on stranded capacity. Nine economies have no penetration value at all, several of them cases in which a facade would be plausible. Coverage is a property of the sources, not a selection, but it bounds what the composite can claim.

**The high-technology indicator distorts at both extremes.** As a share of manufactured exports it is inflated by re-export activity — the Philippine value of 63.98% sets the normalisation

base — and by narrow denominators. It is also blind to digital services trade, which for several economies is the larger channel. A services-inclusive measure holding the denominator fixed is left to future work.

**The analysis is cross-sectional.** Every claim here concerns a single reference year. The facade is interpreted above as a transitional configuration, and the isomorphic-mimicry reading of the typology is a claim about how states respond to assessment over time. Neither can be tested on one wave of data.

**Pakistan’s penetration vintage is unresolved.** Its stranded capacity ranges from 0.413 to 0.218 across the two available figures, moving it from third to fortieth of 184 (Section 4.6). No distributional claim depends on its position, but its individual position should not be quoted without the interval.

**The facade measure is closely related to non-reach.** Stranded capacity inherits most of its cross-country variance from its falsity component, and the two orderings correlate at  $\rho = 0.90$ . We have argued that the conjunction nonetheless does distinct work, on two grounds: it selects a substantially different set of economies at the extremes (Section 3.1), and it survives an external test that a connectivity proxy would fail (Section 3.6). Neither argument makes the two independent, and  $\Phi$  should not be used as though it carried information about state capacity that the EGDI does not already supply. Its content is the co-presence of the two, and only that.

**The typology is classified at the edge of the data’s resolution.** Eighteen per cent of economies have a service–infrastructure divergence smaller than 0.05, which the OSI and TII cannot be expected to resolve. For these the aligned band withholds a configuration rather than asserting one, and no policy reading should be attached to their sign. The threshold itself is a judgement about measurement rather than an estimate, and a different reasonable threshold would move the boundary cases; the compositional results are reported under both readings for that reason.

**The disparity analysis is source-sensitive.** Reported in the supplementary materials, the disparity ordering changes materially according to which modelled informality series is adopted. We report the sensitivity rather than resolving it, and draw no substantive conclusion about inequality from the exercise. Its purpose is to demonstrate that the architecture transfers, and that purpose is served whichever series is used.

## 5.6 Directions for future work

Four extensions follow, and the first two are the consequential ones.

*Time-series application.* The EGDI has been published biennially since 2003 and ITU penetration series run further back still, so stranded capacity can be computed as a panel. This would test the transitional reading of Section 5.3 directly, by asking whether facade economies resolve as reach catches up or persist; and it would discriminate between the isomorphic-mimicry

and relative-cost explanations of the service-side asymmetry, since only the former predicts that the service–infrastructure gap responds to assessment cycles.

*An externally sourced indeterminacy component.* The present indeterminacy is computed within the EGDI. An independent construction would compare assessed provision against measured service uptake — available for OECD and EU members through official statistics on interaction with public authorities, but not for the economies where the facade is largest. Extending such measurement is a data-collection problem rather than a methodological one, and would allow the full triple to be sourced from three distinct instruments.

*Sub-national application.* Following Smolińska-Bryza et al. (2025) for Poland, sub-national estimation would surface heterogeneity that country-level aggregates smooth over — particularly for the large economies that dominate the population in absolute terms.

*Transfer to other indicators.* The architecture is not specific to the digital case. It applies wherever nominal capacity and effective reach diverge, and where the confirming and negating components of a phenomenon are drawn from different instruments. Indicators distinguishing installed capacity from effective operation, such as renewable-energy capacity-utilisation indices, are a natural candidate; the disparity analysis of the supplementary materials is a first demonstration on inequality measurement.

## 6 Conclusion

This paper has developed a neutrosophic decomposition in which state e-government capacity and citizen digital reach are held analytically distinct. The E-Government Development Index is separated into a truth component, a falsity component, and a bipolar indeterminacy component whose poles record opposite misalignments; truth and falsity combine conjunctively to give stranded capacity, and the poles give its configuration. The adjusted score is aggregated with three further digital sub-indicators by geometric mean and combined with the standard health, education, and income dimensions under the imbalance-penalising aggregation the UNDP adopted in 2010. The framework is applied at three coverage tiers set by indicator availability: 184 economies for the facade analysis, 158 for the full composite.

The measure has external support. Conditional on state capacity and income level, stranded capacity predicts lower citizen uptake of digital government services in demand-side household survey data covering 145 economies ( $-0.462$ ,  $p = 2.3 \times 10^{-5}$ ), and predicts nothing about digital payment activity in general. A quantity constructed from two published indices could have been arithmetically transparent and empirically empty; it is not, and it tracks state-mediated uptake specifically rather than connectivity or development.

The facade is widespread, regionally concentrated, and developmentally staged. Intensity is highest in Southern Asia and most prevalent in sub-Saharan Africa, and the regional concentration survives an income control that the North–South partition does not. No economy of the Global North reaches the top quintile of stranded capacity. Within the lower half of the income distribution stranded capacity does not vary systematically, because capacity and reach expand at comparable rates until reach approaches saturation; the poorest economies do not exhibit the largest facades, because they have not yet built the capacity to strand.

The two configurations call for opposite interventions, and their composition is not neutral. Provision-side misalignment is the global norm, but among the economies with the largest facades it inverts: 72% of the top decile is service-side, having built online provision ahead of the network carrying it. Since online provision is assessed from what governments report while infrastructure is computed from telecommunications statistics, the economies furthest from converting capacity into citizen benefit are disproportionately those whose capacity sits in the component that international assessment observes most directly. Whether that reflects strategic response to assessment or the lower cost of building services than networks cannot be settled cross-sectionally, and is the first question a time-series application should address.

The augmentation reorders human development substantially without overturning it, and the reordering is structured: economies whose digital capability runs ahead of their human-development position gain, while small high-income economies dependent on resources or tourism fall, some by more than fifty positions. The ordering is reproduced under alternative aggregators and under full-triple neutrosophic aggregation.

Taken together, these results establish that the gap between what states provide digitally and what citizens can reach is measurable, substantial, and structured — and that its internal configuration carries policy weight the aggregate index conflates. Institutions tracking e-government progress by the EGDI alone will systematically overstate the effective digital capacity of states whose citizen reach lags their investment, and will not distinguish the two categories of shortfall that call for opposite responses. The architecture, holding verified, negating, and uncertain components independent, is not specific to the digital case: it applies wherever nominal capacity and effective reach diverge, or where the confirming and negating components of a phenomenon are drawn from different instruments.

## Data availability

All raw data, the audited analysis panel, and the code required to reproduce every table, figure, and reported statistic in this paper and in the supplementary materials are deposited in a public repository under a persistent identifier: **[PENDING: deposit the archive and insert the DOI and repository name here; the deposit should be made at the revision stage so that the identifier is live at acceptance]**.

The underlying sources are all publicly available. Individual internet use and fixed-broadband subscriptions are drawn from the International Telecommunication Union DataHub ([datahub.itu.int](https://datahub.itu.int)); high-technology exports from the World Bank World Development Indicators ([data.worldbank.org](https://data.worldbank.org)); the E-Government Development Index and its sub-indices from the United Nations Department of Economic and Social Affairs ([publicadministration.un.org](https://publicadministration.un.org)); the standard Human Development Index dimensional indices from the UNDP Human Development Report Office ([hdr.undp.org](https://hdr.undp.org)); the demand-side indicators used in the external validation from the Global Findex Database 2025 ([globalfindex.worldbank.org](https://globalfindex.worldbank.org)); Gini coefficients from the Standardized World Income Inequality Database version 9.x ([fsolt.org/swiid](https://fsolt.org/swiid)); and modelled informal output from the World Bank Informal Economy Database. Retrieval dates are recorded at file level in the deposit and at cell level wherever a carry-forward was applied.

## Competing interests

The authors declare no competing interests.

## Ethics declaration

This study analyses publicly available secondary data at country level. Ethical approval was not required.

## Author contributions

**[PENDING: insert author initials in place of the bracketed placeholders below, and delete this marker]**

[A.B.]: conceptualisation; methodology; formal analysis; validation; writing — original draft; writing — review and editing; supervision; project administration. [C.D.]: data curation; investigation; software; validation. [E.F.]: data curation; investigation; validation. All authors read and approved the final manuscript. The corresponding author had full access to the audited analysis panel and takes responsibility for the integrity of the data and the accuracy of the analysis.

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